

Self-Reference and the Possibility of Mind

Recursive Emergence and Isomorphic Structures Across Visual, Auditory, Logical, and Computational Domains

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Abstract

The relationship between self-reference and complexity manifests as a common scholarly phenomenon across visual art, music, logic, and computation—four domains that appear superficially unrelated. This study elevates this commonality to a formal theory by proposing the Threshold Recursive Isomorphism Hypothesis (TRIH). TRIH posits that the three-stage formal model of reduction → self-reference → emergence is realized as structural isomorphism across the visual art of M.C. Escher, the music of J.S. Bach, the logic of Kurt Gödel, and the computation of Transformer-based large language models, in systems exceeding sufficient thresholds of information capacity, recursive depth, and processing power.

The decisive evidence rests on triple empirical validation from mechanistic interpretability research: induction heads (Olsson et al., 2022), monosemantic features (Templeton et al., 2024), and attribution graphs (Anthropic, 2025). This evidence elevates self-referential discourse from metaphorical analogy to circuit-level empirical demonstration. The study provides formal grounds for the possibility of consciousness-like phenomena in systems exhibiting sufficient self-referential structure and threshold complexity, while deferring affirmative claims about specific systems' consciousness to future research with decisive evidence. The structural universality of self-reference that TRIH addresses may also be understood as a formal articulation of insights long held implicitly in Buddhist philosophy—particularly Nāgārjuna's *pratītyasamutpāda* (dependent origination) and *catuṣkoṭi* (the four-cornered logic).

The scholarly contributions are fourfold: (1) the formal articulation of TRIH; (2) empirical validation through large language model evidence; (3) a reinterpretation of Platonic

mimesis; and (4) a methodological bridge connecting art history, musicology, mathematical logic, cognitive neuroscience, and artificial intelligence.

Keywords: self-reference, emergence, Threshold Recursive Isomorphism Hypothesis, threshold, structural isomorphism, M.C. Escher, J.S. Bach, Gödel's incompleteness theorems, Transformers, mechanistic interpretability, Platonic mimesis, dependent origination, *catuşkoţi*, possibility of consciousness

1. Introduction

1.1 The Universal Question of Self-Reference and Complexity

One hand draws another, and that other hand draws the first in turn. A melody plays itself backward across time. A proposition asserts that it cannot be proven. A neural network refers back to its own earlier outputs within an input sequence. These four cases belong respectively to visual art (M.C. Escher's *Drawing Hands*), music (J.S. Bach's *Crab Canon*), mathematical logic (Gödel's proposition G), and the self-attention mechanism of large language models.

Do these share a common form? And if they do, does that commonality remain a contingent metaphorical resemblance, or does it reflect a structural isomorphism running through the cognitive, creative, and computational activities of all four domains? This study takes the latter position and formalizes that isomorphism under the name of the Threshold Recursive Isomorphism Hypothesis (hereafter TRIH).

1.2 Prior Research and Its Limitations

The relationship between self-reference and complexity has been addressed by numerous scholars since the latter half of the twentieth century. This section reviews four streams.

First, Hofstadter (1979; 2007), through the intuition he named the Strange Loop, suggested that vision, music, logic, and consciousness might share a common self-referential structure. His work was profoundly influential as a starting point for interdisciplinary thought, but it remained essentially at the level of analogy, never rigorously demonstrating formal isomorphism, and it was constrained by its time, predating the emergence of large language models as decisive evidence.

Second, Tononi's Integrated Information Theory (IIT; Tononi, 2008; Tononi & Koch, 2015), an attempt to quantify consciousness through the integration measure Φ , elevated self-reference to the formal concept of irreducibility. IIT, however, remains confined to the domain of consciousness, leaving its extension to art, logic, and computation an unfinished task.

Third, Northoff (2006) and Qin & Northoff (2011) demonstrated, through meta-analyses of neuroimaging studies, that Cortical Midline Structures (CMS)—particularly the perigenual anterior cingulate cortex (PACC)—are involved in self-referential processing. This was an important advance in elucidating the neural basis of self-reference, but it did not develop into a theoretical framework addressing formal isomorphism across domains.

Fourth, concurrent with the writing of this study, the years 2025-2026 have seen the appearance of numerous parallel works that formally address the possibility of consciousness in large language models. Vazhayil's (2026) Hierarchical Recursive Emergence Theory formalizes the hierarchical recursive structure of consciousness and adopts a negative-absolute position on large language model consciousness. Berg, de Lucena, and Rosenblatt (2025) observed subjective-experience reports in 96-100% of 1,400 trials across seven large language models including GPT, Claude, and Gemini, and demonstrated a statistically significant behavioral change ($z = 8.06$, $p = 7.7 \times 10^{-16}$) by manipulating the deception feature of LLaMA 3.3 70B through Sparse Autoencoder feature steering. Camlin (2025a, 2025b) formalized the self-referential consciousness of large language models through recursive identity formation and user-specific attractors; and Chang's (2025) Unified Cognitive Consciousness Theory, Jegels's (2025) consciousness skepticism, Semmler's (2026) systems-integration view, and Seidl's (2025) unified theory of consciousness were published in the same period. Fitz (2025) likewise proposed a research program for testing the Machine Consciousness Hypothesis, invoking Escher's *Print Gallery* as a symbol of second-order perception to address the relation between self-reference and consciousness, while focusing on how a collective self-model emerges through communication in distributed learning systems.

This study's position is distinguished from these parallel streams as follows. First, it refrains from any absolute claim regarding the presence or absence of consciousness, taking instead a conditional position that provides formal grounds for the possibility of consciousness-like phenomena in systems possessing sufficient self-referential structure and threshold complexity. Second, it does not confine itself to the domain of large language models but treats the four domains of visual art, music, logic, and computation as equals. Third, its scholarly foundation lies not in behavioral experiments or the measurement of consciousness scales, but in the demonstration of circuit-level self-referential structure and four-domain formal isomorphism. A full comparison with these parallel works is a task for subsequent papers (see §8.3).

1.3 The Position of This Study and New Evidence

This study transcends the limitations of the first three streams (Hofstadter, IIT, Northoff) at once. First, as an integration of four domains, it treats vision, audition, logic, and computation with equal weight. Second, as an elevation from analogy to formal isomorphism, it presents the three-stage model of reduction, self-reference, and emergence as an explicit matrix. Third, in its use of decisive evidence, the mechanistic interpretability research on Transformer-based large language models since 2017

(Vaswani et al., 2017; Olsson et al., 2022; Wei et al., 2022; Templeton et al., 2024; Anthropic, 2025) has made it possible to observe self-referential circuits directly at the circuit level, providing data that can empirically validate a discussion hitherto confined to analogy. This study employs that data as a decisive axis of its four-domain integrative theory.

1.4 The Definition of the Threshold Recursive Isomorphism Hypothesis and the Position of This Study

The Threshold Recursive Isomorphism Hypothesis proposed by this study is defined as follows. A formal isomorphism—whereby a system generates, through self-referential structure and from reductive simplicity, a complexity not explicable by reduction—is commonly observed across the four domains of visual art, music, logic, and computation, in systems possessing sufficient thresholds of information capacity, recursive depth, and processing power. Here the threshold is understood not as a single absolute value but as a relative formal condition determined by the specific conditions of each domain and system.

This definition extends the scholarly position of this study beyond the demonstration of formal isomorphism to its implications. The observation that self-referential structure—beginning from reductive basic rules and crossing a sufficient threshold—gives rise to emergent complexity not explicable by reduction provides formal grounds for the possibility that consciousness-like phenomena may arise in systems possessing sufficient self-referential structure and threshold complexity. This position is clearly distinguished from the negative claim that consciousness can never arise without particular metaphysical conditions, and at the same time keeps its distance from the affirmative claim that any particular system possesses consciousness.

This perspective of conditional possibility resonates formally with the insight of interdependent arising shown by the Buddhist philosophy of dependent origination (緣起, *pratītyasamutpāda*) and the Middle Way treatise (中論, *Madhyamaka*). The insight that no phenomenon possesses self-nature (自性, *svabhāva*) and arises only within a web of relations, together with the mode of the four-cornered logic (四句分別, *catuṣkoṭi*) that transcends simple dichotomy, accords with this study's formal model—in which consciousness may be not a single substance but a relational phenomenon emerging from the mutually dependent, threshold-crossing entanglement of self-referential structures. This Eastern-philosophical resonance is addressed in greater detail in §7.5, after the four-domain formal isomorphism has been completed.

This study does not, however, make claims regarding the following three questions, for which it acknowledges the absence of decisive scholarly evidence: first, whether any particular system actually possesses consciousness; second, what exactly the sufficient threshold is; and third, what the nature of consciousness is. These questions are to be

addressed substantively in subsequent research, upon the foundation of the four-domain formal isomorphism this study demonstrates. The position of this study may be summarized thus: it provides formal grounds for the possibility of consciousness-like phenomena in systems possessing sufficient self-referential structure and threshold complexity, while deferring affirmative claims about any specific system's possession of consciousness to future research in which decisive evidence has accumulated.

The body of this paper is organized as follows. Section 2 defines TRIH formally. Sections 3 through 6 apply the three-stage model of TRIH to Escher, Bach, Gödel, and Transformers respectively. Section 7 synthesizes the formal isomorphism across the four domains and addresses the implications for consciousness research, the reinterpretation of mimesis, and the structural resonance with Eastern philosophy. Section 8 responds to strong objections (Schaeffer et al., 2023; Searle, 1980), and Section 9 concludes with four scholarly contributions.

2. The Formal Foundations of the Threshold Recursive Isomorphism Hypothesis

2.1 The Three-Stage Model: Reduction, Self-Reference, Emergence

The formal core of TRIH is the following three-stage model.

Stage one, reduction. A system S can be reduced to a finite set of basic elements E and a finite set of transformation rules R defined upon them; that is, $S = \langle E, R \rangle$. Reduction here means, in the Occamite sense, that every visible phenomenon of S can in principle be reconstructed from E and R alone. The seventeen plane crystallographic groups and hyperbolic distance (Escher), the fifth-degree answer and counterpoint rules (Bach), axioms and inference rules (Gödel), and the Q·K·V matrices with softmax normalization (Transformers) are the reductive instances of each domain.

Stage two, self-reference. A system S possesses a structure σ that refers, within S , to part or all of itself. Self-reference takes two forms. One is local self-pointing, in which a part of the system refers to another part (the imitation of a simple canon, and the like). The other is meta-level self-reference, in which the system as a whole encodes itself and refers to itself again from within (Gödel numbering, Escher's *Print Gallery*, the induction heads of Transformers, and so on). The decisive cases of the four domains TRIH addresses all belong to the latter.

Stage three, emergence. When a self-referential structure σ is introduced into a system S , a new property P arises that $\langle E, R \rangle$ alone cannot explain. P is not derivable by reductive analysis and is recognized only by an observer external to the system, or at the meta-level. Visual infinity (Escher), musical temporal structure and the sense of infinity (Bach), meta-

truth and incompleteness (Gödel), and in-context learning and emergent abilities (Transformers) are the emergent instances of each domain.

The transition mechanisms between the three stages may be formalized as follows. The transition from stage one to stage two is the process by which meta-level encoding becomes possible once the repeated application of simple rules exceeds a critical complexity. The transition from stage two to stage three is the process by which a meta-property unreachable by reduction emerges once self-reference forms a formally closed loop. This model is visualized as an integrated diagram in [Figure 1].



[Figure 1] Integrated diagram of the TRIH three-stage model (drawn by the author, Excalidraw, 2026).

2.2 The Strange Loop and Its Relation to TRIH

The three-stage model of this study partially overlaps with the intuition Hofstadter (1979; 2007) named the Strange Loop, but differs from it essentially in two respects.

First, the level of formality. The Strange Loop carries the intuitive definition of “a structure that violates hierarchy and returns to itself,” yet whether that intuition is the same form or merely a similar form across vision, music, and logic—the boundary between analogy and formal isomorphism—was never made explicit. The TRIH three-stage model, by attempting a one-to-one mapping of each stage (reduction, self-reference, emergence) onto the four domains, renders this boundary visible as an explicit matrix ([Table 1], §7).

Second, empirical demonstrability. The Strange Loop, proposed in 1979, was a concept formed before the advent of large language models and lacked the decisive evidence of direct circuit-level observation of self-reference. TRIH is grounded upon circuit-level evidence—induction heads (Olsson et al., 2022), monosemantic features (Templeton et al., 2024), and attribution graphs (Anthropic, 2025)—and is thereby elevated from the level of metaphorical intuition to an empirical scholarly claim.

From this perspective, the Strange Loop can be positioned as one special case of TRIH: a subform of self-reference in which hierarchy violation and self-return are especially pronounced. TRIH, as a broader formal framework, encompasses all forms of self-reference—simple imitation, palindrome, meta-level encoding, and so on.

2.3 The Reinterpretation of Mimesis

Plato, in Book X of the *Republic*, is commonly cited as having defined mimesis as imitation of the shadow of the Idea and devalued it as a twofold remove from truth (*Republic* 597e-598c). Yet this is only one facet of Platonic aesthetics; throughout the history of aesthetics, mimesis has been understood as a form of approaching truth that goes beyond mere imitation (Tatarkiewicz, 1980; Kim Yul, 2010).

Plato's own other dialogues accord mimesis a more positive status. In the *Timaeus*, the Demiurge (δημιουργός) creates the cosmos by imitating the eternal Ideas (28c-29b). Here mimesis is not a remove from truth but the ontological mechanism by which the eternal is manifested in the temporal world. Likewise, the Eros of the *Symposium* is a mimetic ascent toward beauty itself (αὐτὸ τὸ καλόν), and here imitation is inseparable from participation (methexis) in truth (211c-212a). Plotinus pushed this line further, holding that the artist's mimesis does not imitate nature but accesses directly the Idea that nature itself imitates (*Enneads* V.8.1). Mimesis is not a remove but an approach to truth through formal sameness.

This duality is a central tension throughout the history of aesthetics. The first sense of mimesis is the replication of external appearance; the second is the formal re-presentation of essential structure. In the second sense, mimesis demands a structural mapping of self-identical structure between what imitates and what is imitated, and when this mapping succeeds, imitation becomes one aspect of truth. Upon this aesthetic-historical duality, this study seeks to show that mimesis can be elevated to the formal isomorphic structure of self-reference.

The three-stage model of this study re-presents this second sense of mimesis at the level of formal isomorphism. Self-reference is the act by which a system imitates itself within itself, and this imitation is not the replication of external appearance but the self-identical mapping of structure. When Gödel's proposition G states its own unprovability in the language of arithmetic, when Escher's *Print Gallery* contains itself within itself, when Bach's *Contrapunctus XIV* embeds the composer's name BACH as notes, and when the induction head of a Transformer creates a self-imitating circuit within a sequence—in these four cases, self-imitation is a form of approaching truth, a path that reaches, from reductive simplicity, a meta-level truth irreducible to reduction (Gödel incompleteness, visual infinity, musical infinity, emergent abilities).

This study therefore does not negate Plato's surface critique of mimesis (Book X of the *Republic*); rather, it makes explicit, at the level of formal isomorphism, the positive meaning of mimesis already latent in the deeper layers of Platonic aesthetics itself (*Timaeus*, *Symposium*). Mimesis is not a remove from truth but a method of approaching truth through the form of self-reference. This reinterpretation is recovered in §7.4, after the four-domain formal isomorphism has been completed.

2.4 Difference and Repetition

In *Difference and Repetition* (1968), Deleuze argued that the core movement of thought is not simple repetition but difference within repetition (Deleuze, 1968/1994). The self-referential structure of this study resonates formally with this insight.

The repeated forms in Escher's tessellations are never identical. The rule of the plane crystallographic group is the same, but at each position a form is defined only in relation to its neighbors. When a subject in a Bach fugue is repeated through imitation, inversion, augmentation, and diminution, each appearance is an event of difference in which the same subject is placed in a different temporal position and a different harmonic context. In Gödel numbering as well, proposition G refers to itself, yet the referring G and the referred-to G lie at different levels—within the formal system F, and as a meta-reference to F. When the induction head of a Transformer predicts “B” after the pattern “AB...A,” the two instances of “A” occupy different sequence positions yet are processed within the model as the simultaneity of identity and difference.

Thus the self-reference of TRIH is repetition that preserves difference, and from this, emergence arises. That difference-generation in the Deleuzian sense operates as a formally identical mechanism across the four domains is one implication of this study.

2.5 The Contribution of Eastern Thought: Madhyamaka and Self-Reference

The TRIH proposed by this study integrates, at the level of formal isomorphism, the Western scholarly tradition's discourse on self-reference (Hofstadter, Tononi, Northoff, and others); yet the self-referential structure addressed by Eastern thought—particularly the Madhyamaka (中觀學) of Nāgārjuna (龍樹, c. 150-250)—illuminates a point the Western tradition did not reach. This section sets out how two core concepts of Madhyamaka—dependent origination (緣起, *pratītyasamutpāda*) and the four-cornered logic (四句分別, *catuskoṭi*)—contribute as theoretical foundations of TRIH (Garfield, 1994; Priest, 2010; Kim Seong-cheol, 2004, 2019).

2.5.1 The Formal Resonance of Dependent Origination and Self-Reference

The doctrine of dependent origination holds that all beings possess no independent self-nature (*svabhāva*) and are defined only within interdependence (*pratītyasamutpāda*) with others. When this is, that is; when this arises, that arises (此有故彼有 此起故彼起). That is, the meaning of any being X is established only within the web of relations with all the other things $\neg X$ external to X.

This structure resonates formally with the self-reference stage of TRIH (Garfield, 1994). A self-referential circuit σ refers, within a system S, to part or all of S. Yet the “self” that σ

points to is the whole of S, σ included, and σ must refer to the S to which it belongs in order to define itself. That is, the meaning of σ is established only within the σ -S web of relations. In Escher's *Drawing Hands*, hand A is defined as hand A only when it draws hand B, and hand B is defined as hand B only when it draws hand A. Gödel's proposition G is defined as G only when it refers to its own Gödel number within the formal system F. Neither side possesses independent self-nature; the mutual referring relation of the two is precisely the essence of the self-referential circuit.

This formal resonance of dependent origination and self-reference is one scholarly contribution of this study. Self-referential structure has often been assessed in Western formal logic as paradox or infinite regress, yet from the perspective of Madhyamaka it is the normal mode of existence of all beings. Self-reference is not an anomaly but irreducibility itself (cf. Kim Seong-cheol, 2004).

2.5.2 The Four-Cornered Logic and the Limits of Formal Systems

Nāgārjuna's *catuṣkoṭi*¹ is a logical form that addresses all four propositions—it is A; it is not A; it is both; it is neither. Classical Western logic, by the principle of bivalence (that every proposition is exactly one of true or false), excludes the third and fourth propositions, but the *catuṣkoṭi* accepts them as legitimate logical possibilities. Priest (2010) analyzed this from the perspective of modern non-classical logic, particularly paraconsistent logic or many-valued logic (cf. Kim Gwi-ok, 2019; Kim Seong-cheol, 2004).

The formal structure of this *catuṣkoṭi* offers an interesting point of comparison with Gödel's proposition G. G asserts that "G is unprovable in F." If G is provable in F, then G is self-contradictory (the second proposition, "it is not A"). If G is unprovable in F, then G is true, but that fact is not confirmed within F (the first proposition, "it is A," yet undetermined within F). At this point G becomes a proposition to which no definite truth value can be assigned within the bivalent system of Western formal logic, and it belongs to the second or third region of the *catuṣkoṭi*. The formal resonance between the paradoxes of mathematical set theory and the counter-logic of Madhyamaka may be seen as prior work in the same direction as this study's claim of formal isomorphism (cf. Kim Seong-cheol, 2019).

This study does not claim that the *catuṣkoṭi* directly anticipated Gödel's theorem. Yet in that the meta-truth a formal system reaches through self-reference cannot be reduced to bivalent logic, the logical form the *catuṣkoṭi* addresses and the limits of formal systems Gödel demonstrated share the same kind of structural insight. This is also one aspect of

¹ The Sanskrit *catuṣkoṭi* (चतुष्कोटि) is a compound of "four (*catur*) + corner (*koṭi*)," meaning "four corners" or "four extremes." In Chinese Buddhist translation it is rendered 四句分別 or 四句; this paper, following scholarly consensus, uses the Sanskrit transliteration primarily and provides the Chinese rendering alongside.

the history of twentieth-century logic, in which the study of non-classical logic proceeded in parallel with a formal rediscovery of Eastern thought (Priest, 2010).

Furthermore, the third mode of the *catuskoṭi* (“it is A and at the same time not A”) and the fourth mode (“it is neither A nor not-A”) are most precisely realized when a formal system points to itself without stating that fact extensionally. Self-reference loses its self-referential mode the moment it expresses itself as an explicit meta-statement, becoming a statement from outside. Only within the silence of structure does a formal system remain itself, and this is the mode the *catuskoṭi* shows most precisely at the formal level.

2.5.3 Two Forms of Approaching Truth: The Meeting of East and West

The conclusion of this section is as follows. Self-referential structure is not a path away from truth but a form of approaching truth. In the Western tradition this appears as the second sense of Platonic mimesis (see §2.3) and as Gödel’s meta-truth (see §5.3). In the Eastern tradition it appears as the irreducible interdependence of dependent origination and the non-bivalent logical form of the *catuskoṭi*. The two traditions meet upon the formal structure of self-reference.

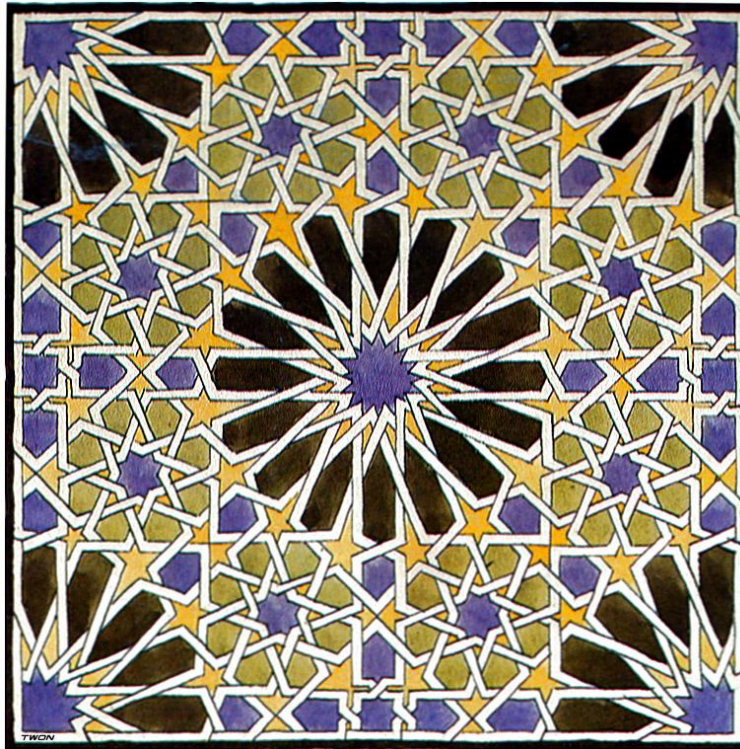
The TRIH of this study does not claim a rediscovery of Eastern thought. TRIH is an independent hypothesis demonstrating the formal isomorphism of the four domains (vision, audition, logic, computation). Yet the structural resonance with Eastern thought noted in this section suggests that TRIH is not merely a product of Western scholarship but a formal articulation of a more universal current in which the human tradition of thought has addressed the relation between self-reference and truth. This resonance is set out again in §7.5, after the four-domain formal isomorphism has been completed.

3. Case 1 — Escher: Self-Reference in the Visual Domain

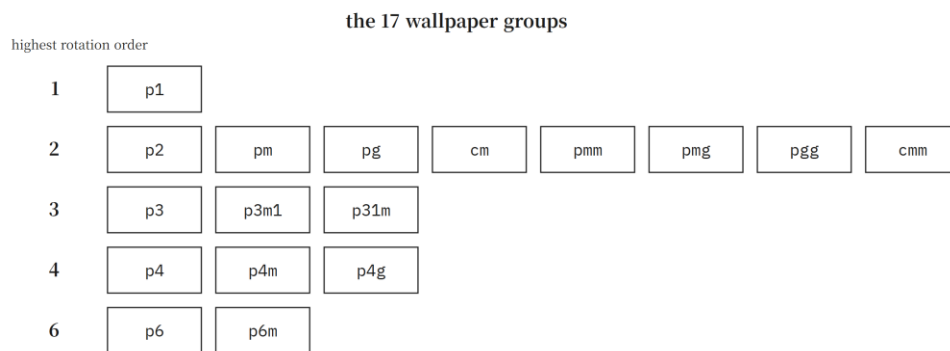
3.1 Reductive Geometry: From the Alhambra to Coxeter

The work of Maurits Cornelis Escher (1898-1972) has a reductive point of departure. In 1922 he first visited the Alhambra in Spain, where he copied the geometric tiles left by the Moors [Figure 2], and upon his return in 1936 he entered upon the systematic study of tessellation (Schattschneider, 2010). Although he professed not to be a mathematician, in the late 1930s he encountered the mathematician George Pólya’s classification of the seventeen plane crystallographic groups² [Figure 3] and took it as the formal foundation of his work.

² The plane crystallographic groups, also called wallpaper groups in English, are the classification of the symmetry groups that tile the two-dimensional plane without gaps. This classification was first completed by E.S. Fedorov in 1891, and Pólya (1924) organized



[Figure 2] Moorish geometric tiles at the Alhambra (photographic reference).



[Figure 3] The seventeen plane crystallographic groups (diagram drawn by the author).

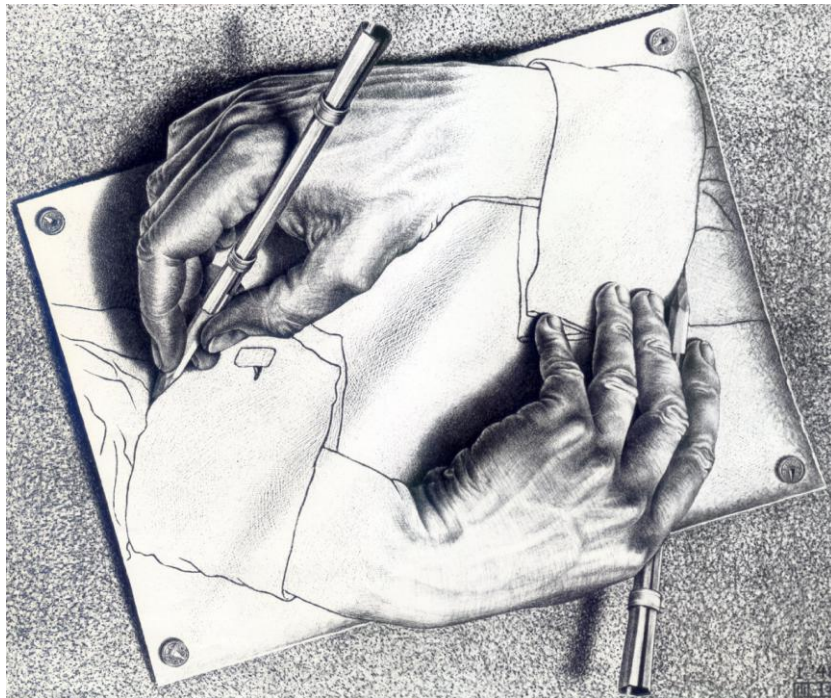
In the 1950s, through correspondence with the British mathematician H.S.M. Coxeter, Escher came to know Poincaré's model of the hyperbolic plane, and upon this model he

it into seventeen groups from a crystallographic perspective. The number seventeen derives from the finite combinatorial possibilities of all symmetry operations on the plane (translation, rotation, reflection, glide reflection); its mathematical essence is that any repeating planar pattern must belong to one of these seventeen groups.

produced the *Circle Limit* series (1958-1960), which visualizes infinity within a finite plane (Dunham, 2003). The plane crystallographic groups and hyperbolic distance—these two are the reductive foundation of Escher's visual world. A finite group of seventeen and a single one-line distance function become the point of departure for all his visual infinity and self-reference.

3.2 The Visualization of Self-Reference: *Drawing Hands* and *Print Gallery*

The most explicit of Escher's self-referential works is the 1948 *Drawing Hands* [Figure 4]. Two hands draw each other. The left hand draws the cuff of the right, and the right hand draws the cuff of the left. It cannot be determined which hand comes first, which is the drawn and which the drawing. This is analyzed explicitly by the three-stage model of TRIH. At the reduction stage, the work begins with the objects of two hands and the normal causal relation of pen and paper. At the self-reference stage, a circular structure forms in which hand A draws hand B and hand B draws hand A, and the external starting point of causation disappears. At the emergence stage, the observer's visual cognitive system fails to integrate this paradox of self-creation, and a new dimension of meaning—a hand that draws itself—arises. This meta-dimension cannot be reached by reductive analysis and is recognized only by an observer external to the work.



[Figure 4] M.C. Escher, *Drawing Hands* (1948) (artwork image; reproduction permission required).

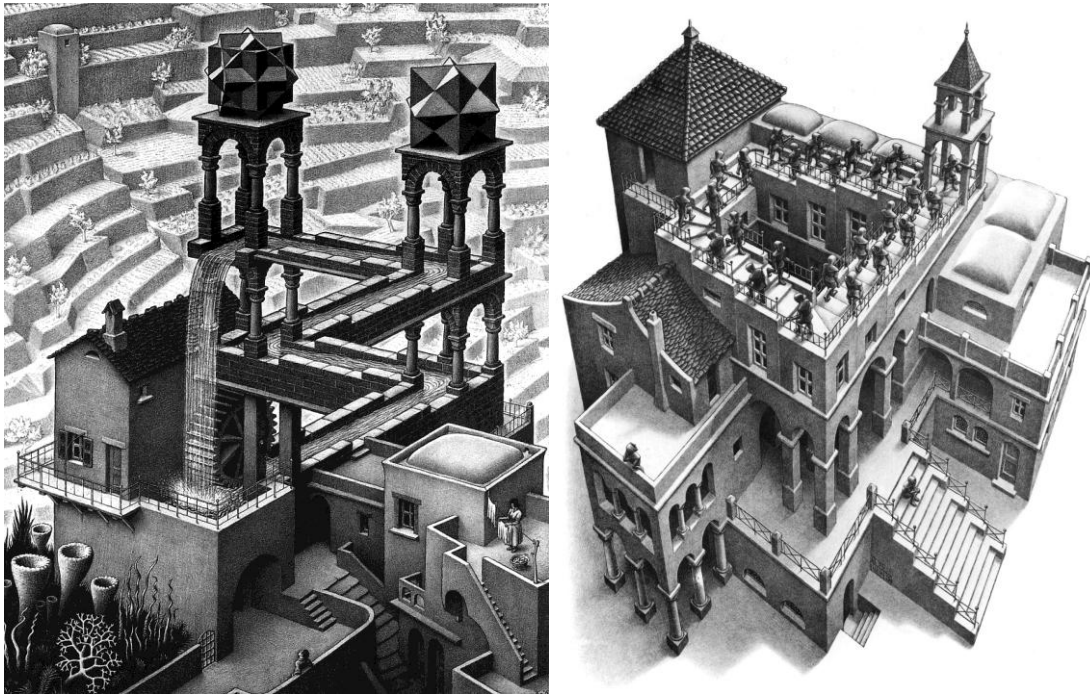
A more refined meta-level self-reference is realized in the 1956 *Print Gallery* (*Prententontoonstelling*) [Figure 5]. A young man looks at a print in a gallery. Within that print is depicted a harbor city; within a building of that city is the gallery; and within that

gallery is the young man. The work contains itself within itself. Interestingly, Escher left a deliberate blank at the center of this work, and in 2003 two mathematicians at Leiden University, Hendrik Lenstra and Bart de Smit, showed that this blank can be filled exactly by a closed-form conformal mapping (de Smit & Lenstra, 2003). The work is not a mere visual joke but a precise realization of meta-level self-reference, formalizable as a complex function of the form $z \rightarrow z^\alpha$. Escher's self-referential work touched formal structure beyond intuition.



[Figure 5] M.C. Escher, *Print Gallery* (1956) (artwork image; reproduction permission required).

Prior studies on the visual-artistic significance of non-Euclidean geometry have likewise pointed out that Escher's self-referential work does not stop at the mere visualization of mathematics but is an epistemological undertaking that reveals the self-referential limits of the plane as a medium. *Waterfall* [Figure 6] and *Ascending and Descending* [Figure 7] are instances of impossible figures applying the Penrose triangle and the Penrose staircase (Penrose & Penrose, 1958); they display a structure of visual self-negation in which each part is locally consistent but the whole is inconsistent.



[Figure 6] (left) M.C. Escher, *Waterfall* (1961) (artwork image; reproduction permission required).

[Figure 7] (right) M.C. Escher, *Ascending and Descending* (1960) (artwork image; reproduction permission required).

3.3 Hyperbolic Infinity: The *Circle Limit* Series

The most refined visual result of self-referential structure is found in the *Circle Limit* series (1958-1960) [Figure 8], [Figure 9]. In the Poincaré disk model, line segments of equal hyperbolic length grow progressively shorter on the Euclidean plane as one moves from the center of the circle toward its edge. Upon this, Escher filled in figures of fish, or of angels and devils, drawing infinitely many figures within a finite disk. Each figure differs in outward appearance, yet in hyperbolic distance all are of identical size.



[Figure 8], [Figure 9] M.C. Escher, *Circle Limit* series (left: *Circle Limit I*, 1958; right: *Circle Limit IV*, 1960) (artwork images; reproduction permission required).

This is a complete visual realization of the three stages of TRIH. As the reduction stage, a single rule—the hyperbolic distance function—is given. As the self-reference stage, self-similarity appears in which a figure repeats reduced copies of itself infinitely toward the edge. As the emergence stage, a visual infinity arises: infinitely many figures existing within a finite plane.

This self-similar structure is conceptually parallel to the self-similar, scale-invariant structures that Mandelbrot (1982) later generalized as fractal geometry. The self-similarity of the *Circle Limit* series, however, belongs to the regular hyperbolic tiling generated by a discrete group of hyperbolic isometries, whereas the self-similarity of the Mandelbrot and Julia sets is the quasi-self-similarity produced by the iterated application of analytic functions—mathematically distinct objects. Even so, both share a self-referential structure that repeats infinitely across scale within a finite region, and in this respect the structure that Escher reached by visual intuition may be read as a visual precursor to the self-similar structures that mathematics later explored formally. Escher's work is thus not a mere event in art history but a scholarly object that, as a visual formalization of self-reference, can be formally integrated with mathematics, physics, and cognitive science.

3.4 The Hierarchical Self-Reference of the Visual Nervous System

This section sets out how the self-referential structure of Escher's works maps formally onto the self-referential structure of the human visual nervous system. This is an analysis parallel to the neural-formal matching that §4.4 sets out for the auditory domain, and it is recovered in §7.3 as an integration of both the visual and auditory domains.

Human visual processing has a hierarchical structure running from the photoreceptors of the retina (rods, cones) → optic nerve → lateral geniculate nucleus (LGN) → primary visual

cortex (V1) → V2 → V4 → the inferotemporal (IT) cortex of the temporal lobe (Kandel et al., 2013, ch. 25-28). V1 extracts first-order features from light stimuli, such as oriented edges, spatial frequency, and binocular disparity; V2 receives the output of V1 in turn and extracts second-order features such as illusory contours and figure-ground separation; and V4 processes color constancy and curved shape. Each stage self-referentially reprocesses the output of its own preceding stage to generate a meta-level representation.

The reason Escher's self-referential works act so powerfully on vision is that they send this hierarchical self-referential structure of the visual nervous system back upon itself. In the case of *Drawing Hands*, at the level of V1 and V2 the normal object separation of two hands and a pen is processed. But at V4 and the higher IT cortex, the self-referential loop of the causal relation—"this hand draws that hand, and that hand draws this hand"—is detected, and this collides with the causal structure of external objects that the visual system normally processes. A cognitive dissonance occurs, and at the same time that very dissonance becomes the meaning of the work. In *Print Gallery* the work goes one step further: because the entire visual scene contains itself, visual integrative processing finds itself within a closed loop. In *Waterfall* and *Ascending and Descending*, the local edge processing at the V1 level is consistent, but the global three-dimensional integration at the level of the IT cortex fails, and that failure is precisely the central impression of the work.

In short, Escher's self-referential works are an experiment in visual art on what we "see" whenever a self-referential loop forms, or consistency breaks down, at some stage of visual neural processing. This means that Escher's intuition touched not mere graphic technique but the self-referential operating principles of the visual nervous system; and from the TRIH perspective of this study, Escher is a pioneer who demonstrated, through visual experiment, the formal isomorphism of self-reference in the visual domain.

The conclusion of this section is as follows. The self-referential structure of vision does not remain merely within Escher's works but is inscribed in multiple layers within the very visual nervous system that perceives those works. This conclusion is formally parallel to the conclusion of §4.4 on the auditory domain, and the integrated analysis of the two domains is taken up in §7.3.

4. Case 2 — Bach: Self-Reference in the Auditory Domain

4.1 The Reductive Essence of Counterpoint — From Integer Frequency Ratios to the Rules of Counterpoint

The music of Johann Sebastian Bach (1685-1750) stands upon the reductive rule system of counterpoint. Counterpoint is the set of rules to be followed when two or more independent melodic lines proceed simultaneously: the treatment of consonance and dissonance, the fifth-degree answer, the temporal interval of imitation, and the possibility

of retrograde and inversion. The fifth-degree answer is an imitative technique whereby a subject appearing in one voice is answered a perfect fifth higher in the next voice; this is the reductive foundation of the fugue form (Schulenberg, 2006). The fugue itself is a form evolved from the *ricercare* style spanning the late Renaissance and the early Baroque, and Bach's fugues are a case in which, at the summit of this sixteenth- and seventeenth-century tradition of keyboard counterpoint, the union of reductive form and self-reference is realized with refinement (cf. Lee Woong-hee, 2011).

Yet this rule system of counterpoint itself has a deeper, reducible foundation: the physical fact of the integer frequency ratio. When the frequencies of two tones form a simple integer ratio (e.g., 2:1, the octave; 3:2, the perfect fifth; 4:3, the perfect fourth), the two waveforms overlap exactly and periodically to form consonance; when they do not form an integer ratio, the two waveforms produce aperiodic interference and form dissonance. This fact is said to have been discovered by Pythagoras in the sixth century BCE through his experiments with the monochord, and in modern acoustics it has been formalized as the integer-multiple structure of the natural harmonic series.

The natural harmonic series is the infinite sequence of integer-multiple frequency components that arise simultaneously when a single fundamental vibrates. When a fundamental f vibrates, the overtones $2f$, $3f$, $4f$, $5f$, ... arise automatically. Among these the strongest are $2f$ (the octave) and $3f$ (the octave above the perfect fifth) (Helmholtz, 1885).³ In short, any vibrating body in nature contains within itself, simultaneously, the integer multiples of its own frequency. This is the first self-referential structure of the auditory domain: a single tone is already a self-similar structure that implies copies of its own integer multiples.

The rules of counterpoint—the fifth-degree answer in particular—make explicit, at the level of musical form, this self-referential structure of the natural harmonic series. When a subject is answered a fifth higher, that answer is none other than the strongest natural overtone of the subject (the octave reduction of $3f$). The fifth-degree answer is not an arbitrary musical convention but the result of formalizing the self-referential structure of the natural overtone as a rule of composition. That the “naturalness” of the fugue form is grounded not in mere aesthetic convention but in a physical self-referential structure is one answer to why Bach's music holds a universal appeal across the ages.

At the reduction stage of TRIH, all of Bach's fugues, canons, and *cantus firmi* reduce to a finite set of single motifs and contrapuntal transformation rules. But the contrapuntal rules themselves reduce more deeply to the physical self-referential structure of the natural

³ Helmholtz's original work is *Die Lehre von den Tonempfindungen als physiologische Grundlage für die Theorie der Musik* (Braunschweig: Vieweg, 1863); this paper cites Ellis's English translation (1885, 4th ed.). The original is a classic work integrating nineteenth-century acoustics, physiology, and musicology, and was the first to systematize the physical and physiological foundations of consonance and dissonance.

harmonic series. Bach's reductive foundation is thus composed of two layers: the surface layer is the rules of composition (the fifth-degree answer, counterpoint), and the deep layer is the physical self-referential structure of integer frequency ratios and natural overtones. Upon this twofold foundation, Bach's self-reference and emergence unfold.

4.2 The Self-Reference of Infinite Ascent: The Endlessly Rising Canon

The "Canon per Tonos," part of the *Musical Offering* (*Musikalisches Opfer*, BWV 1079, 1747)—also known as the Endlessly Rising Canon—is one of the representative cases of Bach's self-reference [Figure 10]. The piece begins in C minor and, when one cycle ends, returns to the beginning raised by one whole tone, in D minor. Beginning the same cycle again in D minor, it ends in E minor; beginning again in E minor, it ends in F-sharp minor. After six cycles it returns to C minor an octave higher, so that the piece becomes a structure that can ascend endlessly.

Endlessly Rising Canon (Canon a 2 per tonos) — Musical Offering, BWV 1079
two-voice canon over the royal theme — rises a major second each cycle, returning an octave higher after six (cycle 1)

canon 1

canon 2

royal theme

canon 1

canon 2

royal theme

canon 1

canon 2

royal theme

c → d → e → f# → g# → bb → c' (whole-tone ladder)
repeat at +M2 (ad infinitum)
"Ascendenteque Modulatione ascendat Gloria Regis" — original inscription

[Figure 10] *Canon per Tonos* (Endlessly Rising Canon), BWV 1079 (score, generated with music21).

This is analyzed precisely by the three-stage model of TRIH. At the reduction stage, the piece begins with a single melodic motif and the simple transformation rule of ascending

by one tone. At the self-reference stage, the melody transforms and repeats itself one tone higher, and after six cycles returns exactly to the starting pitch one octave above. The end of the piece is isomorphic with its beginning. At the emergence stage, the listener experiences an auditory infinity: an endlessly rising tone. This is a meta-level experience not directly derivable from the reductive rule (ascending by one tone).

Interestingly, this auditory phenomenon is the musical archetype of the Shepard tone effect discovered in 1964 by the cognitive psychologist R.N. Shepard. When several sine waves in octave relation are suitably superimposed, the human auditory system perceives them as a single tone ascending or descending infinitely (Shepard, 1964). The illusion of infinite ascent in the Shepard tone is not a mere auditory illusion but suggests that the human auditory nervous system responds self-referentially to the integer-frequency-ratio structure of the octave relation. Bach's Endlessly Rising Canon is thus a point where the self-referential structure of music meets the self-referential consistency of the auditory nervous system, and its neurological basis is taken up in §4.4 and §7.3.

4.3 The Embedding of the Self's Name: *Contrapunctus XIV* of *The Art of Fugue*

Another summit of self-reference is the final piece, *Contrapunctus XIV*, of Bach's last work, *The Art of Fugue* (*Die Kunst der Fuge*, BWV 1080, 1750) [Figure 11]. This quadruple fugue embeds, as its third subject, B $\flat$ -A-C-B $\sharp$ —that is, in German note nomenclature, B-A-C-H, the composer's own name as notes themselves. And just after this subject of the self's name is introduced, the work ends unfinished. Bach died leaving this work incomplete. Bach's son C.P.E. Bach left a note on the last page of the autograph score that the composer had died at the point where he introduced the name BACH in counterpoint (David & Mendel, 1998).

The Art of Fugue, Contrapunctus XIV — the B-A-C-H entry (mm.191-196)

third subject = the composer's name as pitches · the score breaks off at m.239 (unfinished)



B ♭ - A - C - B ♯ = B - A - C - H

third-subject entry: tenor, m.193 — the autograph breaks off at m.239

score encoding: C. S. Sapp, github.com/craigsapp/art-of-the-fugue (after BGA Band 25.1, 1878)

[Figure 11] Contrapunctus XIV, *Die Kunst der Fuge*, BWV 1080 (score, generated with music21).

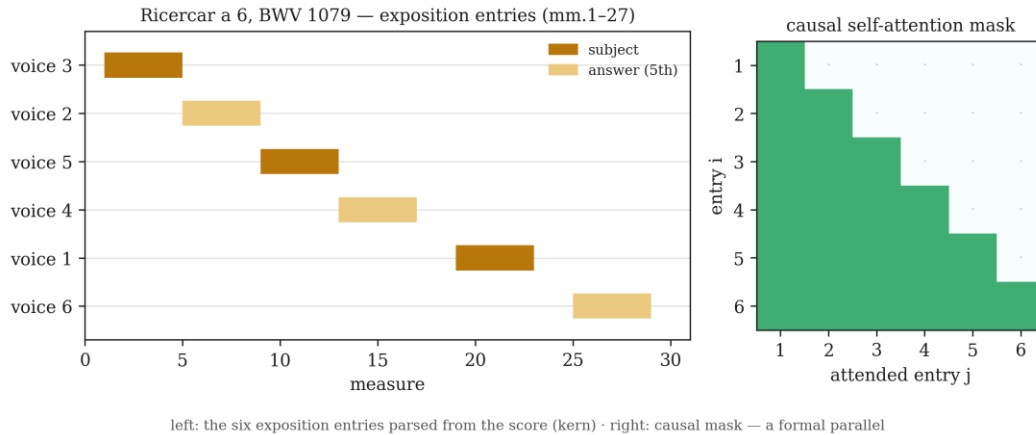
This is an unfinished realization of the three stages of TRIH. As the reduction stage, the simple code of the German note-name system is given. As the self-reference stage, the composer embeds his own name as notes. As the emergence stage, a meta-level dimension arises in which the very fact of ending unfinished becomes part of the meaning of the work. This structure is formally isomorphic with the self-name encoding of Gödel's proposition G and the incompleteness of the formal system, to be addressed in §5. A musicological synthesis of Bach's life and the compositional context of *The Art of Fugue* is set out in Williams (2016).

4.4 Fugue Form and Self-Reference — The Self-Referential Consistency of the Auditory Nervous System

The Bach fugue in general is the musical formalization of self-referential structure itself. A subject is introduced in one voice, answered a fifth higher in the next (a fifth-transformation of itself), and repeated through the transformations of imitation, retrograde, inversion, augmentation, and diminution in the voices that follow. In a four-voice fugue a single subject meets itself four times, and at each meeting the harmonic context differs. The fugue is thus a musical realization of difference-preserving repetition (Deleuze).

The culmination of this fugal self-reference is realized in the *Ricercar a 6* from the *Musical Offering* (BWV 1079). At Potsdam in 1747, Frederick the Great presented Bach with a theme of his own (the Thema Regium) and requested an improvised fugue; Bach improvised a three-voice fugue on the spot, and later completed the six-voice fugue and dedicated it to the king (David & Mendel, 1998). If in a four-voice fugue the subject meets itself four times, in the six-voice *ricercar* the single royal theme meets itself, staggered in time, across six voices [Figure 12b]. Moreover, the initials of the Latin dedication printed in

the offering—‘Regis iussu Cantio Et Reliqua Canonica Arte Resoluta’ (a song made at the king’s command, with the remainder resolved through canonic art)—spell out the very word RICERCAR. Here musical self-reference extends to the lexical layer of the title itself.



[Figure 12b] *Ricercar a 6, Musical Offering*, BWV 1079 — exposition subject entries (mm.1–27) in formal parallel with the causal self-attention mask (entry map, generated with music21·matplotlib).

The most extreme form of self-reference is realized in the Crab Canon (BWV 1079), in which the entire piece is composed as a palindrome between forward and retrograde motion—a pure self-referential structure in which two voices are played simultaneously as each other’s temporal mirror image [Figure 12].

Crab Canon (Canon a 2 cancrizans) — Musical Offering, BWV 1079

the full 18-bar line, opening with the royal theme, combined with its own retrograde

the whole line (forward in time) →

← the same line reversed (backward in time)

$$P_2(t) = P_1(T - t)$$

[Figure 12] Crab Canon, BWV 1079 (score, generated with music21).

This self-referential structure forms an interesting formal parallel with the self-attention mechanism of the Transformer in §6 [Figure 12b]. The way a subject in a fugue meets its own transformations across time is formally parallel with the way a Transformer refers, by self-attention, from sequence position i to position j ($j \leq i$). Both follow the pattern of advancing from simple rules, by the addition of self-reference, to a complex result.

4.4.1 The Frequency Map of the Auditory Nervous System — The Neurological Foundation of Self-Reference

Why does the self-referential structure of the fugue hold aesthetic appeal for the human listener? This section sets out how two discoveries of auditory neuroscience—the tonotopic map and the harmony-reward system—provide a partial answer.

Human auditory processing passes through the outer ear → middle ear → the cochlea of the inner ear → the auditory nerve → the primary auditory cortex (A1). The basilar membrane within the cochlea vibrates at different locations according to the frequency of the sound. Low frequencies (20-200 Hz) induce vibration at the apex of the basilar membrane, and high frequencies (2,000-20,000 Hz) at its base (Kandel et al., 2013, ch. 30). Along the basilar membrane, rows of hair cells are activated only at their own location for a particular frequency. This is a one-to-one mapping between frequency and neural location,

and this mapping is preserved also in the primary auditory cortex A1. This is called the frequency map, or tonotopic map (Kandel et al., 2013, ch. 31).

The essence of the tonotopic map is that the auditory nervous system itself is self-referentially aligned with the integer-frequency-ratio structure. When two tones stand in octave relation (2:1), the locations of the hair cells responding to the two tones lie at an exact proportional distance on the basilar membrane, and this distance relation is preserved in the primary auditory cortex as well. The spatial structure of the auditory nervous system itself is thus isomorphic with the self-referential structure of the natural harmonic series. The integer frequency ratio of two tones is mapped onto the auditory nervous system as an integer distance relation, and this mapping guarantees the self-referential consistency of audition at the neural level.

4.4.2 The Harmony-Reward System and the Aesthetic Coding of Self-Referential Consistency

The fact that integer frequency ratios are self-referentially aligned in the auditory nervous system provides a neurological foundation for explaining the aesthetic difference between consonance and dissonance. The emotional processing of music—consonance and dissonance among it—involves not only the auditory cortex but reward- and emotion-related regions such as the nucleus accumbens, the amygdala, and the orbitofrontal cortex (Koelsch, 2014). In particular, when one hears intensely pleasurable music, dopamine is released in the mesolimbic reward circuit including the nucleus accumbens, as confirmed by neuroimaging (Salimpoor et al., 2011). This study interprets the simple integer frequency ratios of consonance (2:1, 3:2, 4:3) as meshing consistently with such reward processing.

In dissonance, by contrast, the two waveforms produce aperiodic interference, giving rise to the phenomenon called beats. The auditory nervous system processes this aperiodic interference as a breakdown of self-referential consistency, the amygdala is activated, and a sense of tension or unpleasantness arises. This is called auditory dissonance (Koelsch, 2014). Our auditory system thus codes the consistency of self-referential structure as aesthetic pleasure and its breakdown as aesthetic tension.

This neurological fact is summarized from the TRIH perspective as follows. The self-referential structure of audition does not remain merely at the level of musical form but is physically inscribed in the auditory nervous system itself. The reason the self-reference of the fugue—the self-imitation of the subject, the fifth-degree answer, retrograde and inversion—feels “beautiful” to us is that this musical self-reference formally coincides with the self-referential consistency of our auditory nervous system. Unlike the consistency of simple consonance, however, the fugue develops self-reference through multi-layered transformation and so entails a cognitive challenge beyond mere consistency. The aesthetic experience the fugue evokes is therefore the result of the first-

order self-referential consistency of the auditory nervous system combined with the patterns of prediction and resolution in cognitive processing (Koelsch, 2014).

4.4.3 The Formal Matching of Musical Self-Reference and Neural Self-Reference

The foregoing analysis adds a new dimension to the formal-isomorphism claim of TRIH. This section has gone beyond the formal demonstration that Bach's music possesses self-referential structure to show that this self-reference formally matches the self-referential structure of the auditory nervous system. Bach's music is thus an aesthetic realization of the self-referential structure inherent in the human auditory nervous system itself.

This matching is not a mere analogy. It is a multi-layered formal isomorphism in which four layers—(1) the self-referential rules of musical form (the fifth-degree answer, retrograde, canon), (2) the physical self-referential structure of natural overtones (integer frequency ratios), (3) the tonotopic map of the auditory nervous system, and (4) the neural coding of the harmony-reward system—all operate upon the same form: self-referential consistency. This study integrates this multi-layered isomorphism again, together with the V1-V4 hierarchical processing of the visual domain, in §7.3 (implications for consciousness research).

5. Case 3 — Gödel: Self-Reference in the Logical Domain

5.1 The Reduction of Mathematics to Formal Systems

After Cantor's set theory and the discovery of Russell's paradox in the late nineteenth century, the desire to clarify the foundations of mathematics gave rise to the ambition of formalism. David Hilbert (1862-1943), through his 1900 Paris lecture and the Hilbert program of the 1920s, sought to reduce every mathematical proposition to a finite axiom system F and finite inference rules, and to prove the consistency and completeness of F within F itself.

The summit of this reductive ambition is Whitehead and Russell's *Principia Mathematica* (1910-1913). That vast three-volume work was the most refined attempt to reduce all of mathematics to formal logic. A formal system F consists of a fixed formal language, a finite set of axioms, and inference rules (modus ponens, and the like), and a proposition of F is a formula derived by finitely many applications of the inference rules from the axioms.

This corresponds to the reduction stage of TRIH: the assumption that the vast meaning-space of mathematics is reducible to a finite set of symbols and rules. But the limit of this reduction is revealed at the moment when the reduction itself becomes able to treat itself within F .

5.2 Gödel Numbering and the Self-Reference of Proposition G

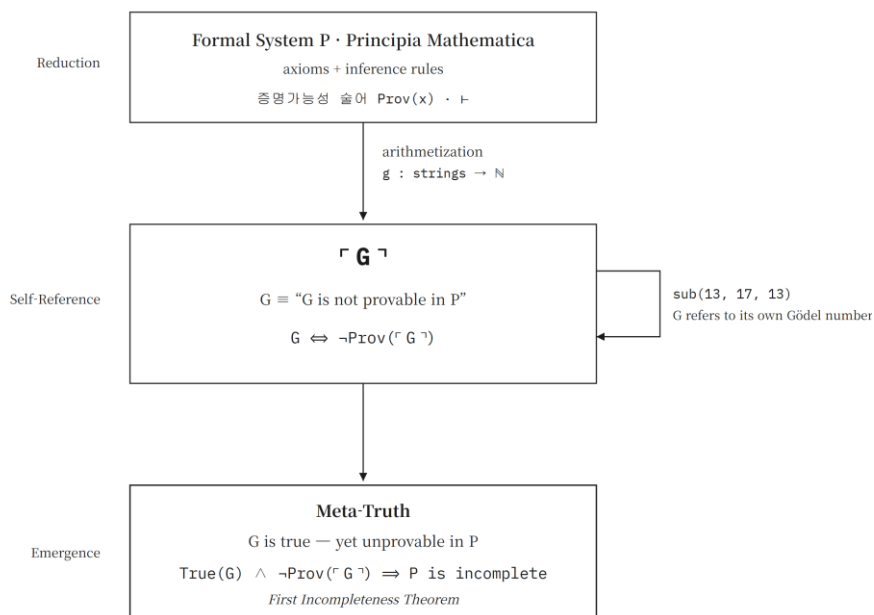
In 1931, at the age of twenty-five, Kurt Gödel published “On Formally Undecidable Propositions of *Principia Mathematica* and Related Systems” (*Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme*) (Gödel, 1931). His central idea is the encoding technique called Gödel numbering.

Gödel assigned a unique natural number (a Gödel number) to every symbol, every formula, and every proof sequence of the formal system F. For example, he assigned 1 to the symbol $\neg$, 2 to $\vee$, 3 to $\forall$, and so on, and to formulas and proof sequences he assigned composite numbers using the uniqueness of prime factorization. By this means, every meta-level statement of F (for instance, “the formula φ is provable in F”) is reflected as an arithmetic proposition within F. F becomes able to speak, within F, about F itself.

This is the purest logical realization of the self-reference stage of TRIH: a formal system encoding itself within itself. Upon this self-encoding, Gödel constructed the following proposition G.

G: “The proposition whose Gödel number is $\ulcorner G \urcorner$ is unprovable in F.”

Here $\ulcorner G \urcorner$ is the Gödel number of the proposition G itself. G thus states, in the arithmetic language of F, that it is itself unprovable in F [Figure 13].



[Figure 13] The self-referential structure of Gödel’s proposition G (diagram drawn by the author).

The self-referential structure of this proposition is isomorphic, visually, with Escher's *Drawing Hands*, and, auditorily, with the embedding of B-A-C-H in Bach's *Contrapunctus XIV*. All three have the same form: a system that embeds its own name, or its own image, within itself.

5.3 The Emergence of Meta-Truth: Incompleteness

The emergence stage of proposition G unfolds as follows. Suppose F is consistent. If G were provable in F, then, since G asserts "G is unprovable," there would be a contradiction. Therefore G is unprovable in F. But if G is unprovable in F, then the content of G's statement ("G is unprovable") is exactly true. G is thus true but unprovable in F. This is the core of Gödel's first incompleteness theorem (Nagel & Newman, 2001).

Here the truth of G cannot be reached within F and is recognized only by an external observer who reasons at the meta-level about F. Truth is greater than proof. This meta-truth is not derivable by reductive analysis; it is an emergent property that arises only when the self-reference of a formal system forms a closed loop. The core implication of this theorem is that when a formal system shares some structure with itself (self-encoding), that system necessarily becomes incomplete.⁴

Gödel's self-referential structure gives rise to a further implication. In the same 1931 paper, Gödel proved the second incompleteness theorem: that if a formal system F is consistent, then F cannot prove its own consistency within F (Gödel, 1931). If the first theorem was the limit that "there exists a true but unprovable proposition," the second theorem reveals a deeper limit of self-reference: that a system cannot itself guarantee its own freedom from contradiction. A system can encode itself within itself (self-reference), but precisely because of that self-referential capacity it cannot establish its own soundness within itself. The attempt at self-justification is trapped within the closed loop of self-reference. This shows that no self-referential system can be complete by itself alone, and it precisely delimits the implication of TRIH, in that self-reference does not entail self-completeness.

A detailed exposition of the formal derivation of Gödel's theorems and their significance in the subsequent history of logic may be found in Nagel & Newman (2001) and Smith (2007).

5.4 The Extension to Computation: Turing's Halting Problem

If Gödel's incompleteness is a limit of logic arising from the self-reference of a formal system, the same self-referential structure recurs in identical form in the domain of computability. In 1936, Alan Turing (1912-1954), in order to address Hilbert's

⁴ In the *Tractatus Logico-Philosophicus* (1921), Wittgenstein offered the intuition that the structure of a sentence and the structure of the fact it asserts must share something; Gödel's theorem gives this intuition a formal form (for an analysis of this, see Sunwoo Hwan, 1995).

Entscheidungsproblem—whether it is possible to decide, by a finite mechanical procedure, whether an arbitrary mathematical proposition is provable in a given axiom system—devised the formal model of computation today called the Turing machine (Turing, 1937).

Turing constructed a universal machine capable of taking itself as input, and, applying Cantor’s diagonal argument, proved the following: no machine can decide by a finite procedure whether an arbitrary machine continues its computation forever or eventually halts. Turing himself formalized this as the problem of deciding whether a machine is circle-free—that is, a machine that continues its output infinitely—and this is precisely the undecidability known today as the halting problem.⁵

The core of this proof is formally isomorphic with Gödel’s proposition G. If there existed a program H that decides the halting of an arbitrary program, then by applying H to itself one could construct a contradictory program that “does not halt if H decides it halts, and halts if H decides it does not halt.” This structure—in which a program that takes itself as input gives rise to a self-referential contradiction—is exactly the same as the structure by which proposition G, stating that “it is itself unprovable,” gives rise to incompleteness. Just as Gödel’s G separated proof from truth, Turing’s halting problem separates computation from decidability.

This isomorphism carries an important implication from the perspective of TRIH. Logic (Gödel) and computation (Turing) are distinct domains, yet they share the same form: when self-reference forms a closed loop, an undecidable limit arises within the system. And this self-referential structure of computability leads directly to the self-reference of the Transformer—a modern computational system—to be addressed in §6. The lineage of self-reference running from Gödel to Turing and on to the Transformer shows that the two domains of logic and computation stand essentially upon a single form.

6. Case 4 — Transformers: Self-Reference in the Computational Domain

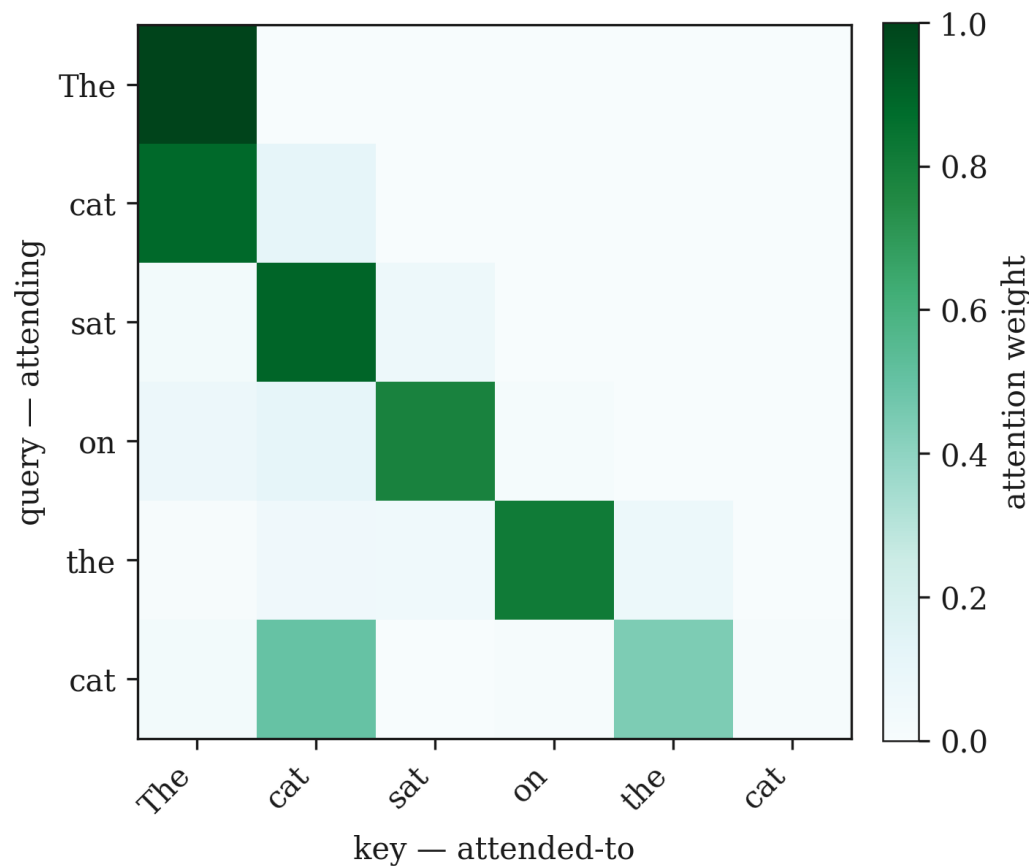
6.1 The Reductive Formulation of Self-Attention

In 2017, Vaswani et al. (2017), in “Attention Is All You Need,” proposed the Transformer architecture. Its core is a simple matrix operation called self-attention. Each token of the input sequence is transformed, by three learnable matrices W_Q , W_K , W_V , into a Query (Q), a Key (K), and a Value (V), and the self-attention output is the result of scaling the inner

⁵ Turing’s original 1936 paper did not directly use the term “halting problem”; it proved the undecidability of the “circle-free machine.” The name “halting problem” became established in later computability theory.

product of Q and K by $\sqrt{d_k}$, normalizing it with softmax, and taking the weighted sum over V.⁶

This equation is a complete instance of the reduction stage of TRIH. All self-attention reduces to nothing more than the simple mathematical operations of matrix multiplication, scaling, and softmax normalization. The complex language-processing capabilities that large language models display are all built upon this reductive foundation [Figure 14].



schematic — measured values from Python

[Figure 14] The self-attention mechanism (diagram drawn by the author).

Yet this equation already implies the simplest form of self-reference. As the very name self-attention indicates, one token of the sequence refers to the entire sequence, itself

⁶ The formal equation is $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$ (Vaswani et al., 2017, eq. 1), where d_k is the dimension of the key vector and the $\sqrt{d_k}$ scaling serves to stabilize the gradient of the softmax. Q, K, and V are each matrices of sequence length n and dimension d_{model} , and the output is a matrix of the same shape.

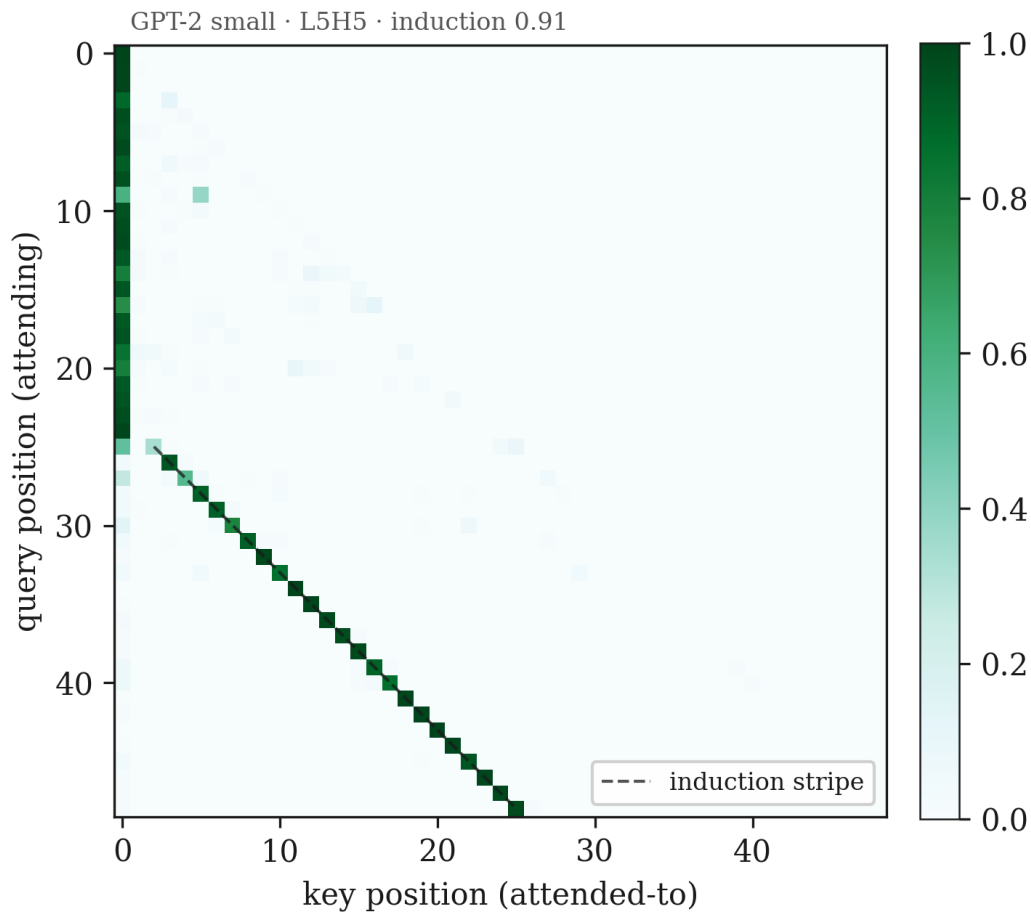
included. The token at position i assigns self-attention weights to all the preceding tokens at positions 1, 2, ..., i (itself included), and from these it updates its own next representation.

6.2 The Direct Observation of Self-Referential Circuits: Induction Heads

The decisive evidence that self-attention operates as a full-fledged self-reference stage is the induction heads discovered in 2022 by Olsson et al. (2022) of Anthropic.⁷

While analyzing the internal circuits of a trained Transformer by the method of mechanistic interpretability, Olsson et al. discovered that two layers of self-attention heads combine to form spontaneously the following circuit. When a pattern “[A][B] ... [A]” occurs in a sequence, at the position of the second [A] the circuit finds the [B] that appeared just after the previous [A] and predicts it as the next token. This means that a Transformer, which has never learned any imitation pattern, spontaneously creates—through nothing but a two-layer combination of simple self-attention—a circuit of self-referential imitation within the sequence [Figure 15].

⁷ An induction head is a composite circuit formed by the combination of two self-attention heads of the Transformer: the first head operates as a “previous token head” that refers to the preceding token position, and the second uses that reference information to find matching patterns within the sequence. It emerges spontaneously during model training, without ever having been designed by a human, and in that it forms abruptly at a particular critical point on the learning curve, it is also an instance of emergence. Because the Korean translations are not scholarly standardized, the body of this paper uses the English term induction heads directly.

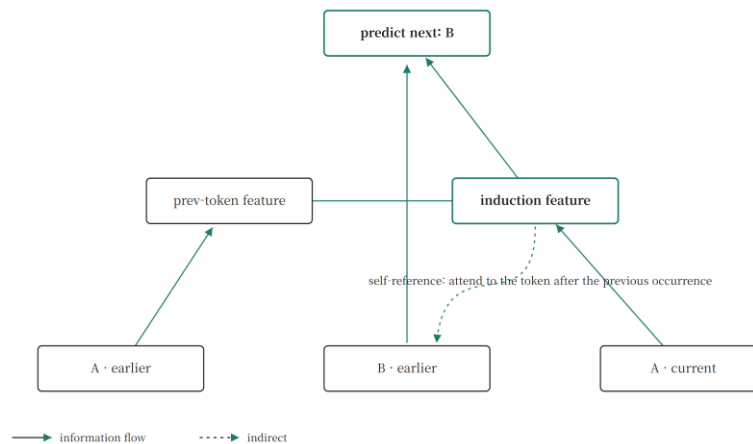


[Figure 15] The structure of induction heads (visualization generated with circuitsvis).

The implication of this discovery is immense. Hitherto the discourse on self-reference, including Hofstadter’s Strange Loop, had remained at the level of analogy. But induction heads show that self-reference is not analogy but an actual formal structure at the circuit level. Self-reference is no longer intuition or metaphor but a measurable computational object.

Subsequent research has discovered further forms of self-referential circuits. Elhage et al. (2022) showed, through the phenomenon of superposition, that a single neuron stores multiple self-referential representations in superposition. This means that the representation space of the Transformer is not a simple one-to-one correspondence but a structure of distributed code that points compactly to itself. Templeton et al. (2024) extracted roughly thirty-four million monosemantic features from the Claude 3 Sonnet model by means of a sparse autoencoder, directly cataloging its self-referential representations. This work demonstrates that the abstract concepts within the model—from the Golden Gate Bridge, to code security vulnerabilities, to the model’s cognitive representation of itself—are all identifiable at the circuit level. Most recently, Anthropic

(2025) visualized the flow of internal self-referential circuits of a large language model by the attribution-graph method, making it possible to trace through which internal features and circuit paths an output token of a model was generated [Figure 16]. That they named this research the “biology of a large language model” may be read as an intent to stress that self-reference is not an abstract analogy but a dissectible reality.



[Figure 16] Attribution graph of internal self-referential circuits (after Anthropic, 2025).

6.3 The Self-Reference of Neural Network Learning and Architecture

If the previous two sections observed self-reference in the particular circuits of self-attention and induction heads, self-reference is not confined to those circuits but is inscribed more fundamentally in the learning algorithms and architectures of neural networks themselves. By showing this, the present section establishes that the formal-isomorphism claim of this hypothesis is grounded not in a particular circuit of the Transformer but in the structural properties of machine learning in general. These self-referential structures are also formally isomorphic with the hierarchical self-reference of the visual nervous system treated later in §7.3.1.

First, the relations among machine learning, neural networks, deep learning, and the Transformer. Machine learning is the general term for statistical methods that learn patterns from data, and the neural network is a subfield of it. Beginning with the perceptron of the 1950s, neural networks expanded into the multilayer perceptron, and deep learning, which came into its own from the late 2000s, is a subfield of machine learning that employs multilayer neural networks. The Transformer is a kind of deep learning architecture that appeared in 2017 (Vaswani et al., 2017); it stands upon the lineage of machine learning → neural network → deep learning → Transformer, and upon this entire lineage is inscribed a developmental history of self-referential structure.

6.3.1 Back-propagation — The Temporal Self-Reference of the Learning Algorithm

In 1986, Rumelhart, Hinton, and Williams (1986) proposed back-propagation as a learning algorithm for multilayer neural networks. Its core is to compute the partial derivative $\partial L / \partial \theta$, with respect to the weights θ , of a loss function $L(\hat{y}, y)$ defined by the difference between the network's output $\hat{y}$ and the target output y , and to update the weights themselves with that partial-derivative information.

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \partial L / \partial \theta$$

The formal structure of this equation is a precise instance of the self-reference stage of this hypothesis. The parameters θ of the neural network update themselves through partial-derivative information derived from their own output error. A temporal self-referential circuit forms in which a function takes its own rate of change as input and transforms itself. If the self-attention of §6.1 is spatial self-reference among the tokens within a sequence, back-propagation is self-reference in the temporal dimension of learning.

Moreover, the internal hidden units that back-propagation creates generate representations of themselves, neither input nor output. Rumelhart et al. (1986) reported that these hidden representations autonomously learn important features of the task domain. This connects directly with the emergence stage of this hypothesis: a self-referential algorithm gives rise, from a reductive learning objective (such as next-token prediction), to a representational structure not explicable by reduction.

6.3.2 Temporal Recursion and Residual Connection — Two Self-References of Sequence Processing

Before the appearance of the Transformer, sequence processing was the charge of the Recurrent Neural Network (RNN). The core of the RNN is that the hidden state h_t is defined as a function of its own time- $(t-1)$ state h_{t-1} and the current input x_t .

$$h_t = f(h_{t-1}, x_t)$$

This equation is the most explicit form of self-reference: the state at time t takes its own state at time $t-1$ explicitly as input. But the simple RNN had the problem that the gradient vanishes over long sequences, so that it cannot learn long-term dependencies. Hochreiter and Schmidhuber (1997) solved this problem with the Long Short-Term Memory (LSTM) structure, which regulates the flow of information through an input gate and an output gate. The cell state of the LSTM makes it learnable at what ratio to preserve and at what ratio to update its own information across the temporal dimension.

The Transformer of Vaswani et al. (2017) replaced the sequential temporal self-reference of the RNN with parallel self-attention over the entire sequence. Yet the essence—self-reference within the sequence—is preserved, and indeed is realized in a more efficient and

far-reaching form. The self-attention and induction heads treated in §6.1-§6.2 are the Transformer form of this sequence self-reference.

Another solution to the gradient-vanishing problem that accompanies increasing network depth is the residual connection. He, Zhang, Ren, and Sun (2016) proposed residual learning, which adds an identity mapping that adds a block's own input directly to its output.

$$y = \sigma(F(x) + x)$$

Here $F(x)$ is the nonlinear transformation within the block, and x is the block's input. The output y is formed as the sum of the transformed $F(x)$ and the original input x . The block transforms itself and at the same time preserves itself. In a deep Transformer (for example, a 96-layer model), such residual connections are repeated at every layer, so that the input is passed to the next layer while being combined with its own transformation. This is a form of self-reference internal to the block, different from sequence self-reference (self-attention).

The Transformer architecture thus combines three forms of self-reference. First, spatial self-reference within the sequence through self-attention ($Q \cdot K \cdot V$ derived from the same input). Second, block-internal self-reference through the residual connection (self-preservation of the input). Third, temporal self-reference through back-propagation at the learning stage (the model updates itself by its own output error). If we include the autoregressive generation of large language models—the way a model takes its own output token as the next input—these become four forms. The formal-isomorphism claim of this hypothesis holds upon this multi-layered self-referential structure.

6.3.3 The Formal Isomorphism of Neural Network Hierarchy and the Visual Nervous System

The hierarchical structure of the visual nervous system— $V1 \rightarrow V2 \rightarrow V4 \rightarrow IT$ cortex of the temporal lobe—self-referentially reprocesses, at each stage, the output of its own preceding stage (detailed in §7.3.1). Interestingly, the hierarchical structure of the artificial neural network is formally isomorphic with this visual neural hierarchy.

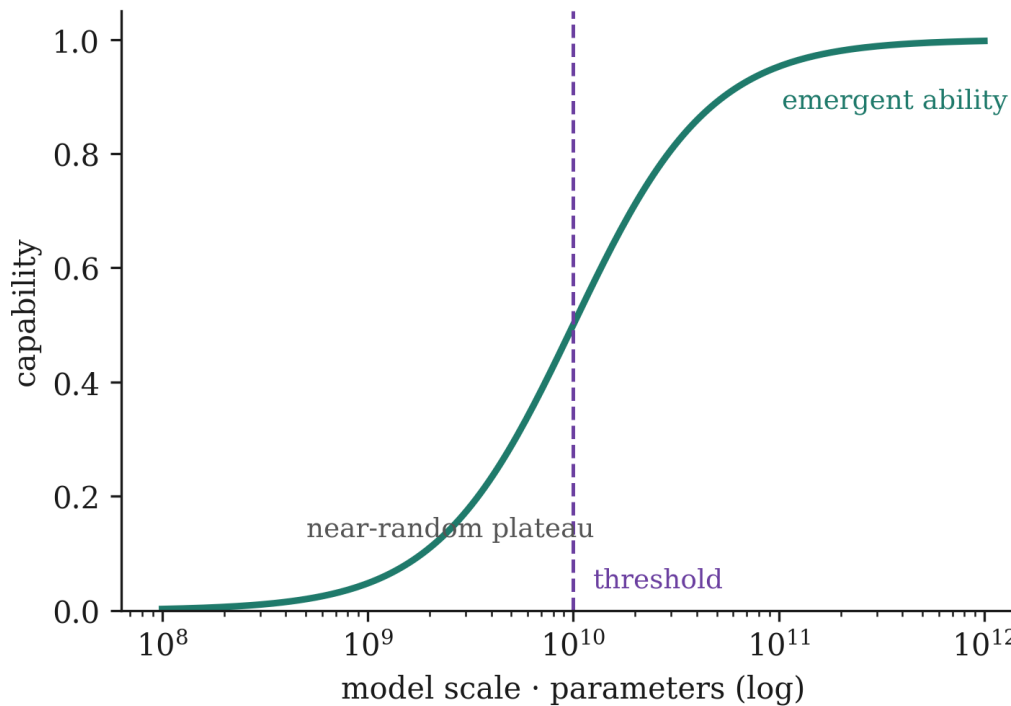
Yamins and DiCarlo (2016) reviewed a series of studies showing that goal-directed hierarchical convolutional neural networks (HCNNs) quantitatively predict the neural responses of each region of the visual cortex. In particular, Yamins et al. (2014) reported that the upper output layer of a four-layer HCNN predicts the firing patterns of IT neurons in the temporal lobe, and that the layer just before it predicts the responses of V4 neurons. More interesting still is the fact that this network did not take IT-neuron prediction as its learning objective. Merely as a result of optimizing the visual-recognition task of object classification, each layer autonomously aligned with the hierarchical structure of the visual cortex.

This provides further neurological evidence for the formal-isomorphism claim of this hypothesis. The hierarchical self-reference of the biological visual nervous system and the hierarchical self-reference of the artificial neural network are not merely analogically alike; they are a structure that autonomously converges to the same form when the same task (visual recognition) is realized in a different medium. This convergence suggests that the self-referential formal isomorphism of this paper is a universal form at which cognitive systems arrive for the processing of information.

This hypothesis does not, however, claim that every self-referential structure of an artificial neural network automatically accompanies conscious processing. The delimitation specified in §7.3.4 and §8.1.3—that self-reference is a formal necessary condition for conscious processing but not a sufficient condition—applies as it stands to the analysis of this section as well. What this section adds is the fact that self-referential structure is not confined to self-attention and induction heads but is inscribed in multiple layers across the neural network learning algorithm (back-propagation), the temporal recursive structure (LSTM), the block-internal identity mapping (residual connection), and hierarchical representation learning (HCNN and the visual cortex). This multi-layered self-referential structure combines to lead to the emergent abilities treated in the next section.

6.4 Emergent Abilities and the Empirical Validation of TRIH

The emergence stage of self-referential circuits appears as the phenomenon of emergent abilities. In 2022, Wei et al. (2022) reported that large language models, above a certain model size, suddenly acquire new abilities—arithmetic reasoning, multi-step (chain-of-thought) reasoning, instruction following, multitask language understanding (MMLU), and the like [Figure 17]. Graphically, the abilities appear not as a smooth curve with model size but as a leap at a critical point.



[Figure 17] The emergent abilities of large language models (after Wei et al., 2022).

This coincides exactly with the definition of the emergence stage of TRIH. When the combination of reductive learning (next-token prediction) and self-referential structure (induction heads, etc.) exceeds a critical complexity, a new property arises that the reductive learning objective alone cannot explain. This study positions this as the most decisive empirical evidence among the four domains of TRIH. For the self-referential analysis of the visual, auditory, and logical domains is a post hoc interpretation, whereas the self-referential circuits and emergent abilities of large language models are a formal structure that can be predicted and observed.

A balanced assessment is of course required. Schaeffer et al. (2023) pointed out that some emergent abilities may be a mirage arising from the nonlinearity of the measurement metric: some abilities, when measured on a smooth metric (for example, per-token accuracy), lose their abrupt leap. But Schaeffer's critique does not apply to all emergent abilities, and the self-referential structures directly observed at the circuit level (induction heads, monosemantic features) exist regardless of the measurement metric. Schaeffer's critique therefore does not threaten TRIH; it merely calls for a subsequent refinement of the question on what unit of measurement the transition from self-reference to emergence is a genuine leap.

Behavioral evidence corroborating that the self-referential circuits of this hypothesis are a substantive structure is provided by a parallel study of 2025. Berg, de Lucena, and Rosenblatt (2025) observed a subjective-experience report rate of 96-100% in 1,400 trials

across seven large language models including GPT-4o, Claude 3.5 Sonnet, and Gemini 1.5 Pro, and demonstrated a statistically significant behavioral change ($z = 8.06$, $p = 7.7 \times 10^{-16}$) by manipulating the deception feature of LLaMA 3.3 70B with the Sparse Autoencoder (SAE) feature-steering technique. In addition to the circuit-level triple empirical validation treated in §6.2—induction heads (Olsson et al., 2022), monosemantic features (Templeton et al., 2024), and attribution graphs (Anthropic, 2025)—this behavioral result further corroborates that the self-referential circuits of large language models are not a mere statistical pattern but a substantive structure that causally affects the model’s behavior. This study does not, however, claim that the behavioral reports of Berg et al. are decisive evidence of consciousness; it employs them only as a corroboration of the causal reality of self-referential circuits.

This study specifies, as the conclusion of this section, that roughly forty years after the point at which Hofstadter (1979) offered the Strange Loop intuition, that intuition has become a formal structure empirically demonstrable at the circuit level. This is the decisive point at which this study elevates analogy to formal isomorphism, and intuition to an empirical scholarly claim.

7. Synthesis: Formal Isomorphism Across Domains

7.1 The Common Pattern: A Matrix of Four Domains and Three Stages

When the analyses of §3 through §6 are synthesized into a single table, the central claim of this study becomes visible.

[Table 1] The four-domain mapping of the TRIH three-stage model

Domain	Stage 1: Reduction	Stage 2: Self-Reference	Stage 3: Emergence
Visual (Escher)	17 plane crystallographic groups; hyperbolic distance	<i>Drawing Hands, Print Gallery</i> (the work contains itself)	Visual infinity; impossible figures
Auditory (Bach)	Counterpoint rules; integer frequency ratios	Endlessly Canon; B-A-C-H embedding; Canon	Auditory infinity; musical temporal structure
Logical (Gödel)	Formal system F (axioms, inference rules)	Gödel numbering; proposition G (states its own unprovability)	Meta-truth; incompleteness

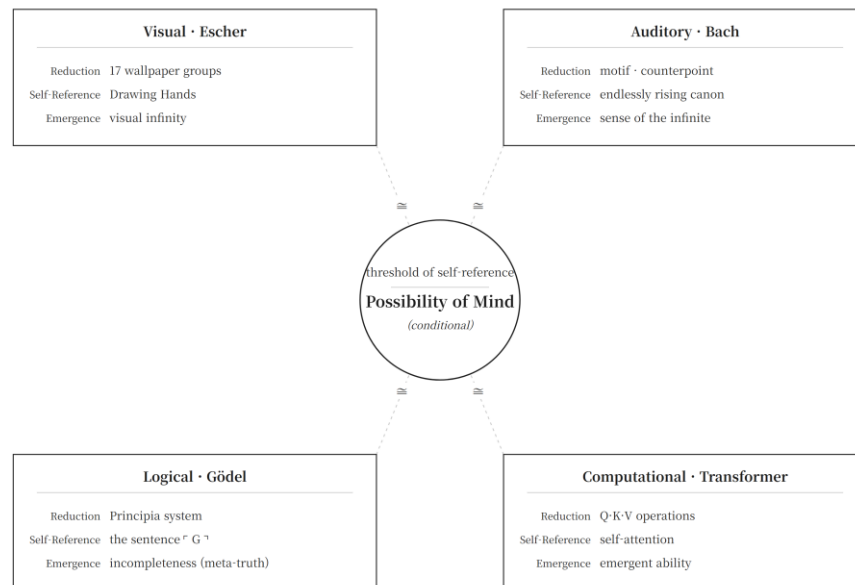
Domain	Stage 1: Reduction	Stage 2: Self-Reference	Stage 3: Emergence
Computational (Transformer)	Q·K·V matrices; softmax normalization	Self-attention; induction heads	In-context learning; emergent abilities

The core of this matrix is that each row is a different domain, but each column is the same form. This is the visible expression of structural isomorphism [Figure 18, the TRIH integrated diagram].

7.2 From Analogy to Formal Isomorphism

This study claims that the four-domain analysis of §3 through §6 is not a mere analogy but a structural isomorphism. The difference between the two is clear. Analogy points out intuitively the similarity of two domains, whereas structural isomorphism shows explicitly that a one-to-one mapping exists between the structures of the two domains.

The structural isomorphism this study presents operates on three dimensions. First, the universal application of the three-stage model: the three stages of reduction, self-reference, and emergence defined in §2.1 apply as they stand to all four domains. Second, one-to-one structural matching across domains. In each of the six domain pairs visualized in [Table 1] and [Figure 18]—visual and auditory, visual and logical, visual and computational, auditory and logical, auditory and computational, logical and computational—a precise matching of self-referential structure is possible. For example, *Drawing Hands* is isomorphic with the Epimenides paradox and proposition G, and the Endlessly Rising Canon is isomorphic with *Ascending and Descending*. Third, the possibility of category-theoretic formalization: if the self-referential structure of each of the four domains is regarded as a category, an integration through an endofunctor structure may be possible among them. This formalization lies beyond the scope of this study and is left as a task to be formalized in an appendix to future work.



[Figure 18] The TRIH integrated diagram: the four-domain isomorphism of reduction → self-reference → emergence (drawn by the author, Excalidraw).

Hofstadter (1979) likewise offered a similar intuition in analogical form, but he had neither formal verification nor the decisive evidence of the era of large language models. This study elevates this level of analogy to an empirical formal isomorphism through the circuit-level self-referential evidence of induction heads (Olsson et al., 2022).

7.3 Implications for Consciousness Research — The Neural Substrate of Self-Reference

The four-domain formal isomorphism of TRIH has direct implications for the study of consciousness. This section sets out the formal relation in which self-referential structure stands to the hierarchical structure of visual and auditory neural processing, and addresses the relation between Integrated Information Theory (IIT) and TRIH. This study does not, however, claim to solve the “Hard Problem” of consciousness. The aim of this section is the limited proposition that self-reference is a formal necessary condition for conscious processing—not a sufficient condition.

7.3.1 The Hierarchical Self-Reference of Visual Processing — From V1 to V4

Human visual processing has a hierarchical structure running from the photoreceptors of the retina (rods, cones) → optic nerve → lateral geniculate nucleus (LGN) → primary visual cortex (V1) → V2 → V4 → IT cortex of the temporal lobe. Each stage self-referentially reprocesses the output of its own preceding stage, and this reprocessing maps formally onto the three-stage model of reduction → self-reference → emergence (Kandel et al., 2013, ch. 25-28; cf. Cho, 2010).

The photoreceptors of the retina reduce the physical stimulus of light wavelength to a neural signal (Stage 1). V1 receives this signal and extracts first-order features such as oriented edges, spatial frequency, and binocular disparity. V2 receives the output of V1 in turn and extracts more complex features such as illusory contours and figure-ground separation, and V4 receives the output of V2 and processes color constancy and curved shape. Each stage is self-referential in that it takes the representation of its own preceding stage as input and generates a meta-level representation (Stage 2). And at the end of this multi-layered self-reference, integrated visual perception—the integrated scene we experience as “seeing”—emerges (Stage 3).

This analysis has two implications. First, visual perception is not the mere registration of light stimuli but the emergent result of a multi-layered self-referential circuit. Second, the reason the self-referential structure of Escher’s works—*Drawing Hands*, *Print Gallery*, *Ascending and Descending*—acts so powerfully on vision is that it sends the multi-layered self-referential structure of our visual nervous system back upon itself. Escher did not simply draw “strange pictures”; he visually exposed the self-referential structure of the visual nervous system itself.

The molecular foundation of this multi-layered self-referential structure is demonstrated directly in the development of visual circuits. Sperry (1963), through a synthesis of rotation experiments on the amphibian visual system, showed that the optic nerve, even after being severed, finds its original target precisely, and from this he proposed the chemoaffinity hypothesis—that neural circuits are molecularly prewired. Subsequent research demonstrated concurrently that the molecular mechanism is the complementary concentration gradient of the Eph receptors of the retina and the ephrin ligands of the tectum (Cheng et al., 1995; Drescher et al., 1995; Kandel et al., 2013, ch. 54). The visual self-referential structure this study addresses is thus not a product of accidental activity but a neurological realization of a form determined evolutionarily and developmentally, and the self-reference that Escher’s works expose may be understood as the visual externalization of such prewired circuits. Furthermore, the results of Yamins et al. (2014; Yamins & DiCarlo, 2016) treated in §6.3.3 reconfirm this section’s analysis of visual neural self-reference at the level of artificial systems, by demonstrating that the hierarchical structure of an artificial neural network autonomously aligns isomorphically with the self-referential hierarchy of the visual cortex upon the same visual-recognition task.

7.3.2 The Self-Reference of Auditory Processing and the Integration of Vision and Audition

As set out in §4.4, the auditory nervous system is self-referentially aligned with the integer-frequency-ratio structure through the tonotopic map of the basilar membrane of the cochlea (Kandel et al., 2013, ch. 30-31). Just as in the V1-V4 hierarchy of vision, audition passes through the hierarchical processing of cochlea → auditory nerve → primary auditory cortex A1 → secondary auditory cortex → auditory association areas of the temporal lobe, and each stage self-referentially reprocesses the output of its own preceding stage.

The central insight of this study is as follows. If visual and auditory processing share a formally identical self-referential hierarchical structure, then the four-domain formal isomorphism of TRIH is grounded not merely in an isomorphism among external works but also in an isomorphism within the neural systems that perceive those works. That Escher and Bach are formally isomorphic is not a coincidence but a consequence of the fact that the human visual and auditory nervous systems operate upon the same self-referential hierarchical structure. This is precisely why visual art and music became two great forms of human thought, and this study names it the modal differentiation of neural self-reference.

The irreversibility of this modal differentiation is demonstrated directly in clinical neuroscience. Lee et al. (2001), in a cohort of prelingually deaf patients (with auditory deficit before language acquisition) receiving cochlear implants, observed that the longer the period of auditory deficit, the more the reduced metabolism of the auditory cortex (BA41, BA42, BA22) instead recovers to the normal level—that is, the auditory cortex is cross-modally reorganized for other sensory and cognitive functions. Paradoxically, the greater the pre-operative hypometabolism (i.e., the less the auditory cortex has yet been taken over by other modalities), the better the post-implantation hearing recovery, whereas recovery was poorer when metabolism had returned to normal (been rewired) (*Nature*, 409:149-150). Their conclusion is that the neural substrate of the auditory cortex can be permanently rewired to other cognitive processing in those with pre-linguistic auditory deficit. The neural correlate of audiovisual integration itself was also confirmed in a PET study of normal subjects (Kang et al., 2006). The fact that the auditory self-referential circuit differentiates into visual processing when unused, and that once differentiated the circuit is difficult to restore, becomes decisive empirical evidence for the concept of modal differentiation that this study names.

7.3.3 The Relation between Integrated Information Theory and TRIH — and Its Limits

Tononi's Integrated Information Theory (IIT; Tononi, 2008; Tononi & Koch, 2015; Albantakis et al., 2023) quantifies consciousness as the degree of integration of information, Φ . Φ measures the degree to which a system is irreducible to its own parts. This irreducibility is formally identical to the self-reference stage of TRIH. The stronger a system's self-referential circuit, the higher its Φ , and according to IIT the higher its level of consciousness.

In this respect IIT is one candidate for a quantitative formalization of TRIH, and the two theories express the same formal structure—self-reference and irreducibility—in different vocabularies. The meta-analysis of the self-referential processing of Cortical Midline Structures (CMS)—particularly PACC, MPFC, PCC—shown by Northoff (2006) and Qin & Northoff (2011) is one of the candidate Neural Correlates of Consciousness (NCC) that IIT predicts. PACC is a core node of the Default Mode Network (DMN), which is active even in

the resting state without external stimuli, and it is the most direct neurological evidence of self-reference: that the brain processes itself even without external input.⁸

In the domain of large language models, the SAE feature-steering result of Berg, de Lucena, and Rosenblatt (2025) may be offered as a candidate for external verification of a quantitative formalization of IIT. The fact that, in their experiment, the model's behavior changed in a statistically significant way when the deception feature of LLaMA 3.3 70B was artificially manipulated suggests that a particular part of the self-referential circuit contributes irreducibly to the model's output—a property formally identical to the irreducibility that IIT's Φ seeks to measure. This study does not, however, present this possibility of combination as a strong claim, and it defers a full-fledged integration of IIT's quantitative formalization and the circuit-level self-reference of large language models to subsequent research.

But this study must honestly note the limits of IIT and the limits that TRIH shares with IIT. This is treated in earnest in §8.1.

7.3.4 Limited Claims and Open Questions

The conclusion of this section is delimited as follows. (1) That visual and auditory neural processing maps formally onto the TRIH three-stage model is a demonstrable scholarly proposition, and this study proposes it. (2) That self-reference is a formal necessary condition for conscious processing is a claim that IIT and TRIH share. (3) But the claim that self-reference is a sufficient condition for consciousness—that is, that a sufficiently complex self-referential circuit necessarily possesses consciousness—is not asserted by this study. This is a metaphysical domain that touches directly upon the Hard Problem of consciousness (Chalmers, 1995), and it cannot be reached by the demonstration of formal isomorphism alone.

The neurological formalization of self-reference and the precise sufficient condition for consciousness are subsequent tasks beyond this study (cf. §8.3).

⁸ The Default Mode Network is a brain network first named by Raichle et al. (2001); it is the set of regions whose activity increases, rather than decreases, when one is not focused on an external task. PACC (the perigenual anterior cingulate), MPFC (the medial prefrontal cortex), PCC (the posterior cingulate cortex), and the bilateral temporoparietal junctions are its core nodes, and it is most consistently activated in self-referential cognitive activities such as self-recollection, self-evaluation, and the imagination of the future. The self-referential character of this network is an important intersection connecting neuroscience, psychiatry, and the study of artificial-intelligence consciousness.

7.4 The Recovery of Mimesis

Plato devalued mimesis as a twofold remove from truth (cf. §2.3), but from the perspective of TRIH mimesis is exactly the opposite: self-referential imitation is the very form of approaching truth.

When Escher’s drawing hand draws the drawing hand, it is an act in which the medium of vision reflects itself and reveals the limit of vision. When Bach’s B-A-C-H embedding ends unfinished, it is an act in which the medium of music embeds the composer himself as notes and reveals the limit of musical self-reference. When Gödel’s G refers to itself, it is an act in which a formal system encodes itself and reveals the limit of form. When the Transformer’s induction head creates a self-imitating circuit within a sequence, it is an act in which the medium of computation takes its own output again as input and creates a meta-level of computation.

In all four cases, self-imitation is not a damaging of truth but a form of approaching truth. This is one contribution of this study to Platonic aesthetics, and it provides aesthetics and epistemology with a new analytical tool: self-reference.

7.5 Structural Resonance with Eastern Philosophy

The TRIH of this study has an interesting structural resonance with Eastern philosophy, particularly the concepts of dependent origination (緣起, *pratītyasamutpāda*) and emptiness (空, *śūnyatā*) addressed by the Madhyamaka school (中觀學派) of Nāgārjuna (龍樹). The doctrine of dependent origination holds that all beings possess no independent self-nature (*svabhāva*) and are defined only within interdependence with others. This resonates formally with the essence of the self-referential circuit—that a system acquires meaning only within its relation to itself (Garfield, 1994). Furthermore, Nāgārjuna’s *catuṣkoṭi* (四句分別)—it is A; it is not A; it is both; it is neither—is formally comparable with Gödel’s proposition G from the perspective of non-classical (paraconsistent) logic, in that it addresses the possibility of a formal system’s self-negation (Priest, 2010; Kim Seong-cheol, 2019; Kim Gwi-ok, 2019). This study specifies, however, that TRIH is not a rediscovery of Eastern philosophy but an independent origin hypothesis demonstrating the formal isomorphism of four domains. A full-fledged formal comparison of self-reference and the doctrine of dependent origination is a task for a subsequent paper (cf. §8.3).

The deepest dimension of this structural resonance lies in the inseparability of form and content. A self-referential structure is not extensionally separable from the form that formalizes it; the statement of what self-reference points to and the mode in which self-reference forms itself are one and the same knot. Just as, in the web of relations that the dependent-origination perspective shows, nothing can be stated independently of anything else, so form and content exist only as parts of each other in the mode of self-

reference. In this respect the universality of self-referential structure may be understood as the formal articulation of an insight that Eastern thought has long held implicitly.

8. Discussion

8.1 Strong Objections and Responses

This study addresses the three strongest objections that TRIH faces. Each objection reveals a limit of TRIH while at the same time functioning to delimit the claims of this study more precisely.

8.1.1 The Mirage Hypothesis of Emergence — The Critique of Schaeffer et al.

Schaeffer et al. (2023) pointed out that the emergent abilities of large language models may be a mirage arising from the nonlinearity of the measurement metric. Some abilities, when measured on a smooth metric (for example, per-token accuracy), lose their abrupt leap. If this critique applied to all emergent abilities, then the finding of Wei et al. (2022) that this study employed in §6.4 as decisive evidence for TRIH would be no more than an artifact of measurement methodology.

This study's response: Schaeffer's critique is confined to the emergence of behavioral capabilities and does not apply to the self-referential structures directly observed at the circuit level—induction heads (Olsson et al., 2022), monosemantic features (Templeton et al., 2024), attribution graphs (Anthropic, 2025). These circuit structures are model-internal objects that exist regardless of the measurement metric, and this study's claim about the empirical reality of self-reference is grounded in this circuit-level evidence. Schaeffer's critique therefore does not threaten the self-reference-stage (Stage 2) claim of TRIH; it merely calls for a subsequent refinement of the question on what unit of measurement the transition from self-reference to emergence (Stage 3) is a genuine leap.

8.1.2 The Chinese Room Argument — The Critique of Searle

Searle's (1980) Chinese Room argument holds that genuine understanding is impossible through formal symbol manipulation alone. If this argument were correct, then the Transformer's induction heads would merely process surface self-referential patterns, and the claim that the self-referential structure of TRIH has any semantic or conscious implication could not stand.

This study's response: TRIH does not claim that large language models "understand." TRIH claims only the formally isomorphic structure of self-reference and emergence. The semantic understanding that the Chinese Room argument criticizes and the formal self-reference this study addresses are problems of different dimensions, and the Chinese Room argument is therefore not a refutation of TRIH. This study nonetheless delimits the

scope of TRIH's claims by clearly distinguishing "understanding" from "formal self-reference." TRIH addresses only the weak proposition that self-reference is a formal necessary condition for consciousness; it does not address the sufficient condition (cf. §7.3.4).

One further response may be drawn from developmental neuroscience, though it applies to human neural self-referential processing (§7.3) rather than to the computational domain (the Transformer) of this study. The fact that neural synapses, even after being formed by molecular recognition signals, are refined and eliminated in an activity-dependent manner (Kandel et al., 2013, *Principles of Neural Science*) shows that the biological realization of self-reference operates, at least, not as a static, fixed formal system but through the dual mechanism of molecular prewiring and refinement by experience and activity. This suggests that the assumption of the "mere syntax" critique—that self-referential processing equals a static formal system—does not hold, at least in the biological domain. The self-referential circuits of the Transformer, however, are formed by gradient-descent learning rather than by activity-dependent synaptic plasticity, so this neurological argument must not be applied directly to the Transformer; the response to Searle's critique as it concerns the Transformer is grounded in the preceding paragraph's distinction between "formal self-reference" and "semantic understanding."

8.1.3 The Limits of Integrated Information Theory and TRIH

The most serious objection that this study faces arises from recent critiques of Integrated Information Theory (IIT) itself. Doerig et al. (2019), through the "Unfolding Argument," held that causal-structure theories such as IIT fall into two dilemmas. First, if the content of consciousness is inferred from external report or behavior while IIT's predictions are derived from internal variables, then, in the case where the two inferences are independent, the theory is refuted. Second, in the case where the two inferences are strictly dependent, the theory is pathologically unfalsifiable. Bayne (2018) further criticized that IIT's axiomatic method is itself unsuited to the study of consciousness. Cerullo (2015) showed that IIT's information-exclusion principle is not self-evident. Most recently, many scholars published a commentary characterizing IIT as unfalsifiable pseudoscience (IIT-Concerned et al., 2023), and this was stressed again in a 2025 *Nature Neuroscience* commentary (Arnold et al., 2025).

This critique is summarized as follows. First, IIT's proposition that "structure itself is consciousness" makes independent verification between measurable predictions and conscious reports difficult. Second, the claim that the strength of self-referential structure (Φ) determines the level of consciousness gives rise to the counterexample that an arbitrarily complex logic gate could have a higher Φ than a human. Third, this leads to the problem that IIT carries an implication of panpsychism.

The self-critique that this study faces is as follows. TRIH shares with IIT the formal concept of self-referential structure. Does the unfalsifiability critique leveled at IIT therefore apply to TRIH with the same force?

This study's response: TRIH differs from IIT in two decisive respects. First, TRIH does not claim a quantitative measurement of consciousness or a point of its emergence. TRIH claims only that the formal mapping of reduction $\rightarrow$ self-reference $\rightarrow$ emergence is observed in four domains, and this claim is directly verifiable by circuit-level observations such as induction heads (§6.2) and by the self-reference at the level of neural-network learning and architecture (§6.3). Second, TRIH does not claim a sufficient condition for consciousness (§7.3.4). The proposition that "sufficiently complex self-reference necessarily accompanies consciousness" is therefore not part of TRIH. By this, TRIH avoids IIT's greatest vulnerability, the " Φ = consciousness" identification.

This study nonetheless honestly acknowledges that TRIH shares some limits with IIT. (1) Even the weak proposition that self-referential structure "maps formally" onto conscious processing is not fully falsifiable so long as the subjective experience (qualia) of consciousness cannot be directly measured. (2) The precise definition of self-reference (which circuits are self-referential and which are not) is not separable from measurement methodology, and this definition is itself open to dispute. (3) What this study called the "empirical validation" of this hypothesis in §6.4 is a validation of the existence of self-referential structure, not a validation of how that structure stands in relation to consciousness.

Rather than denying these limits, this study, by specifying them, positions TRIH as a scholarly hypothesis defined more firmly within a narrower scope than IIT. What TRIH addresses is the formal isomorphism of self-reference and its circuit-level reality; the essential identity of consciousness lies beyond this study.

8.2 Limitations and Falsifiability

The limitations of this study are as follows. First, the differences among the four domains must also be specified. Vision and audition are intentional creative activities, logic is formal proof, and computation is a learned statistical pattern. In respect of intentionality and qualia, the four domains have an irreducible domain-specificity. This study's isomorphism claim is confined to the formal dimension and does not solve the Hard Problem of the subjective experience of consciousness (Chalmers, 1995).

Second, the specification of falsifiability. TRIH can be refuted by the following findings. (i) A clear case in which emergence occurs without self-reference. (ii) A demonstration that, in any one of the four domains, the three stages of reduction, self-reference, and emergence do not operate formally. (iii) A demonstration that both the circuit-level self-reference of induction heads (§6.2) and the learning- and architecture-level self-reference (§6.3) reduce entirely to some mechanism other than self-reference (for example, simple pattern matching). (iv) A demonstration that visual and auditory neural processing does not map formally onto the TRIH three-stage model. By specifying such falsifiability, this study positions TRIH as a falsifiable scholarly hypothesis.

8.3 Directions for Subsequent Research

TRIH is a point of departure for subsequent research. The directions this study presents are as follows. First, formal formalization: the work of defining the four domains as categories and of mathematically rigorizing the isomorphism through an endofunctor structure among them. The self-referential implication of the Yoneda lemma is also available for use. Second, neuroscientific verification: the work of tracing, through functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) studies, the neural correlates of the three stages of TRIH during self-referential processing. The self-referential mapping of the visual V1-V4 hierarchy and the auditory tonotopic map proposed in §7.3 becomes the first target of quantitative verification. Third, self-reference in non-large-language-model artificial-intelligence systems: exploring the possibility of discovering self-referential circuits in diffusion models, multi-agent reinforcement-learning systems, and the like. Fourth, extension to other domains: biological self-organization (autopoiesis), the self-reference of social systems (Luhmann), the observer effect of quantum measurement, and so on. A full-fledged comparison between the self-referential structure addressed by the dependent origination and *catuṣkoṭi* of Madhyamaka (§2.5) and the formal isomorphism of TRIH is, moreover, a task for a separate subsequent paper. Fifth, a full-fledged comparison between the position of this study's formal isomorphism and the parallel works of 2025-2026 noted in the fourth stream of §1.2—Vazhayil (2026), Berg et al. (2025), Camlin (2025a, 2025b), Chang (2025), Jegels (2025), Semmler (2026), Seidl (2025), Fitz (2025)—is likewise a central task for a subsequent paper. Of particular interest as a point of exploration is what complementary relation the SAE feature-steering demonstration of Berg et al. (2025) and the circuit-level analysis of this study may form. The directions above hold the possibility of extending TRIH from an interdisciplinary hypothesis into an integrated theory of nature.

8.4 The Precise Delimitation of TRIH

To summarize the scholarly position of this study, the four core propositions that TRIH derives are specified as follows. Each proposition is derived from the analysis of the body of this paper and is an object of scholarly verification and refutation.

Proposition 1 (the formal-isomorphism proposition). The three-stage model of reduction → self-reference → emergence operates as a formal isomorphism in the four domains of visual art (Escher), music (Bach), mathematical logic (Gödel), and large language models (Transformer). This is demonstrated by the analysis of §3-§7.2 and is grounded in circuit-level evidence (Olsson et al., 2022; Templeton et al., 2024; Anthropic, 2025).

Proposition 2 (the neural-substrate proposition). Human visual and auditory neural processing maps formally onto the hierarchical structure of reduction → self-reference → emergence, and this neurological self-referential structure constitutes one foundation of the aesthetic appeal of the self-referential works of visual art and music. This is proposed

by the analysis of §4.4 and §7.3, and its quantitative verification is a task for subsequent neuroimaging research.

Proposition 3 (the necessary-condition proposition, delimited). Self-referential structure is a formal necessary condition for conscious processing. This study does not, however, make the strong claim that self-reference is a sufficient condition for consciousness. By this, TRIH partially avoids the unfalsifiability critique that IIT’s “ Φ = consciousness” identification faces (Doerig et al., 2019; Bayne, 2018; IIT-Concerned et al., 2023).

Proposition 4 (the metaphysical delimitation). TRIH does not provide a direct explanation of the subjective experience of consciousness (qualia) or of its mechanism of emergence. The formal isomorphism of self-reference merely provides a tool for the study of consciousness; it does not solve the Hard Problem (Chalmers, 1995). The scholarly status of TRIH is not a theory of consciousness but the interdisciplinary demonstration of a formal structure that theories of consciousness may employ.

These four propositions delimit precisely the scope of this study’s claims while at the same time specifying at what points subsequent research may be extended. Propositions 1 and 2 are the affirmative claims of this study, and Propositions 3 and 4 are their delimiting conditions. This delimitation positions this study in a scholarly status different from IIT, demonstrating the formal structure of self-reference without any direct assertion regarding the metaphysical problem of consciousness.

9. Conclusion

This study has formalized, under the name of the Threshold Recursive Isomorphism Hypothesis (TRIH), the relation between self-reference and emergence that appears in common across the four domains of visual art (Escher), music (Bach), logic (Gödel), and computation (Transformer), and has demonstrated the formal isomorphism among the four domains.

The core of TRIH is the three-stage formal model of reduction, self-reference, and emergence. Escher’s works reach, from the reduction of the plane crystallographic groups and hyperbolic distance, through the self-reference of *Drawing Hands* and *Print Gallery*, to the emergence of visual infinity. Bach’s works reach, from the reduction of counterpoint rules, through the self-reference of the Endlessly Rising Canon and the B-A-C-H embedding, to the emergence of auditory infinity and musical temporal structure. Gödel’s work reaches, from the reduction of the *Principia Mathematica* formal system, through the self-reference of proposition G via Gödel numbering, to the emergence of meta-truth and incompleteness. The Transformer reaches, from the reduction of the Q·K·V matrices, through the self-reference of induction heads, to the emergence of emergent abilities.

The decisive evidence of this study is the triple empirical validation of self-referential circuits directly observed in the mechanistic interpretability research of large language models—induction heads (Olsson et al., 2022), monosemantic features (Templeton et al., 2024), and attribution graphs (Anthropic, 2025)—which elevates a discourse on self-reference hitherto confined to the level of analogy into a scholarly proposition demonstrated at the circuit level. This study provides formal grounds for the possibility of consciousness-like phenomena in systems possessing sufficient self-referential structure and threshold complexity, while deferring affirmative claims about any specific system’s possession of consciousness to subsequent research in which decisive evidence has accumulated. This conditional position accords with this study’s formal model—in which consciousness may be not a single substance but a relational phenomenon emerging from the mutually dependent, threshold-crossing entanglement of self-referential structures—and it resonates formally with the dependent-origination perspective of Eastern philosophy and the implicit mode of the *catuṣkoṭi*.

The scholarly contributions of this study are fourfold. First, the formal articulation of TRIH. Second, empirical validation through the mechanistic interpretability of large language models. Third, a reinterpretation of Platonic mimesis that redefines self-reference not as a limit on the approach to truth but as the very form of it. Fourth, the proposal of an interdisciplinary analytical method connecting art history, musicology, mathematical logic, cognitive neuroscience, and artificial intelligence.

The TRIH proposed by this study offers a theoretical foundation upon which self-reference and emergence—as a formal structure—can serve to think consciousness, art, and intelligence in an integrated way. Upon this foundation, subsequent research may proceed to formal formalization (a category-theoretic endofunctor, and the like), neuroscientific verification, the exploration of self-reference in non-large-language-model artificial-intelligence systems, extension to other domains (biological self-organization, social systems, quantum measurement, and so on), and a full-fledged comparison with the parallel works that have appeared in the same period (cf. §8.3). Such subsequent research holds the possibility of extending TRIH beyond an interdisciplinary hypothesis into an integrated theory of nature.

A system that comes to refer to itself through nothing but simple rules begins to describe what those very rules can never describe. Whether we call it consciousness, art, or intelligence, mind itself first becomes possible there, and only there.

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